Finite Element Analysis of Second-Order Strain-Gradient Elasticity with Ellipsoidal Nonlocal Characteristic Lengths

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Abstract

This study presents an anisotropic strain-gradient elasticity framework in which direction-dependent microstructural interactions arise from a geometrically defined nonlocal domain. The resulting characteristic-length tensor comes from the second moments of a three-dimensional ellipsoidal averaging domain, giving it a direct geometric meaning similar to a fabric tensor. A positive-definite strain-gradient energy is discretized using smooth, continuously differentiable Hermite finite elements in one, two, and three dimensions, together with a corrected consistent mass formulation. For an axial characteristic length of 0.20 m and transverse lengths of 0.02 m, rotating the microstructural orientation produces a monotonic response. Static displacement increases by about 4%, while the fundamental frequency decreases from 186.3 Hz to 173.1 Hz. The normalised frequency ranges from 1.053 to 0.979 relative to a volume-equivalent isotropic reference of 176.8 Hz. The axial length dominates the transverse contribution, producing a 5.8% stiffness increase compared with only 0.5% in the transverse-dominant case. A Directional Stiffening Index, measured against a volume-equivalent isotropic reference, is predominantly positive, ranging from 0% to about 30% across the anisotropy ratios and orientations studied. Only a few high-angle configurations show small negative values. These trends allow the stiffness and vibration frequency of anisotropic microstructured materials to be tailored through controlled variation of characteristic length and orientation.

Keywords: Strain-gradient elasticity; anisotropic nonlocality; ellipsoidal averaging neighbourhood; characteristic length tensor; C^1-continuous finite elements; size-dependent stiffening.

1. Introduction

Classical elasticity provides an efficient continuum description when the characteristic structural dimension is much larger than the underlying microstructural scale. Its constitutive relation is local: stress depends on strain at the same material point, and no intrinsic length parameter appears in the governing equations [1]. Classical continuum modelling and numerical analysis remain the working basis for a wide range of engineering problems, for example wear of processed steels [2], difference schemes for nonlinear evolution equations [3], panel-method aerodynamics [4], shear banding in crushable soils [5], fracture of architected cubane structures [6], plates on orthotropic foundations under moving loads [7], metallic glasses [8] and sandwich plates under slamming loads [9].

However, it is restrictive for composites with microstructures, thin films, additively manufactured lattices and mechanical metamaterials, where the presence of pores, fibres, grains, inclusions and/or architected unit cells affect the apparent stiffness, vibration response, wave dispersion and boundary-layer behaviour [10, 11]. In these materials, a continuum

model thus needs to contain characteristic lengths; these lengths should be able to take into account the directionally organized microstructure.

This is overcome by generalized continuum theories, which extend the kinematics or constitutive response. Theories of couple-stress and higher order elasticity were developed to incorporate measures of curvature, moment stresses, strain gradients, and higher order boundary conditions [12–15]. Eringen-type nonlocal elasticity, on the other hand, viewed stress as an average response over a neighbourhood of material points [16–18] in parallel. The two are closely related: nonlocal averaging gives a physical interpretation to long-range interactions, while gradient approximations transform weak nonlocality into differential equations that can be solved numerically on a boundary value problem. In practice, however, they are not very useful because the general higher order constitutive tensors are too complicated, the nonlocal kernels are ambiguous, and consistent boundary conditions are hard to enforce on finite domains.

Within this framework, the simplified strain-gradient elasticity, notably the Aifantis form, provides a numerically more efficient alternative. These models regularize localization, eliminate singularities, and predict size-dependent responses by inclusion of a small internal length parameter [19–21]. Experimental and computational studies have confirmed the relevance of strain-gradient effects in microscale deformation and dynamic problems [22, 23]. Discrete-to-continuum and homogenization-based arguments further show that gradient terms may emerge naturally from lattice or representative-volume descriptions [24, 25]. Micromorphic constructions and unifying variational frameworks have likewise been used to embed strain-gradient and stress-gradient elasticity within a common formulation [26, 27]. Yet most simplified models use a scalar length scale, implicitly assuming that nonlocal interactions are isotropic. This assumption is inadequate for layered media, fibre-reinforced solids, anisotropic crystals, and architected lattices, where the interaction range may differ substantially from one direction to another.

Anisotropic gradient elasticity has therefore become an important extension of scalar-length models. Tensorial characteristic lengths allow the weak nonlocal response to depend on direction, and this idea has been developed through separable anisotropic gradient, internal-length tensor formulations, and symmetry classifications of three-dimensional strain-gradient elasticity [28–30]. These studies establish that the internal length is not necessarily a scalar material constant; it may be a second-order tensor carrying information about microstructural orientation. However, in many formulations this tensor is introduced phenomenologically or through homogenized coefficients. The link between a geometrically defined nonlocal averaging domain and the resulting anisotropic length tensor remains less explicit, although such a link is essential for parameter interpretation and model calibration.

Recent work reinforces the same need from different perspectives. Physically based nonlocal strain-gradient models have related kernel functions and intrinsic lengths to polymer-network descriptors [31], while second-order homogenization of three-dimensional lattice materials has shown that strain-gradient continua can reproduce size effects that classical homogenization misses [32]. Recent developments in non-classical continuum theories have expanded applications to smart structures and multiphysics. Nonlocal strain gradient models have been applied, mostly within isogeometric analysis, to magneto-electro-elastic, porous and functionally graded nanoplates [33–36], to graphene-reinforced composite nanoscale plates [37] and to honeycomb sandwich nanoscale plates [38]. Nonlocal elasticity has also been implemented through boundary element [39] and nonlocal operator [40] methods. These studies confirm that internal length parameters significantly modify stiffness, vibration and wave dispersion, motivating anisotropic extensions where length itself becomes tensorial. Hierarchical lattice studies further indicate that higher-order stiffness tensors improve static and dynamic predictions in architected solids [41]. At the defect scale, nonlocal simplified strain-gradient elasticity has clarified the joint role of Eringen-type and Aifantis-type lengths in regularizing dislocation fields [42]. Related nonlocal multiphysics studies, including the recent work of Gupta and co-workers on piezo-thermoelastic, piezo-semiconductor, voided, and fractional Moore–Gibson–Thompson media, show that nonlocal parameters can significantly modify wave speed, attenuation, vibration response, and particle motion in smart small-scale systems [43–46].

Although these developments broaden the scope of nonlocal and gradient modelling, they also highlight a remaining limitation: most formulations are either analytical and geometry-specific, homogenization-dependent, or not directly connected to a unified finite element framework for anisotropic finite-domain problems. Finally, the numerical treatment

of strain-gradient elasticity represents an additional challenge. In a primal (displacement-based) Galerkin formulation, fourth-order governing equations necessitate C^1-continuous approximation. The mixed and staggered finite element methods relax the continuity requirement but add new fields, stabilization choices or a saddle-point structure [47–49].

The direct C^1 formulations preserve the variational structure, and when appropriate interpolation functions exist, they are also convenient to apply for general gradient elasticity. For example, such approximations can be achieved in a classical sense via cubic Hermite interpolation, Bogner–Fox–Schmit elements and isoparametric Hermite constructions [50, 51]. Recently, finite element developments for elasticity with microstructure, three-dimensional C^1 gradient elasticity and associated verification benchmarks [52, 53], as well as virtual element formulations [54, 55] have elucidated the computational requirements of higher-order continua. A unified one-, two-, and three-dimensional C^1 implementation coupled with anisotropic characteristic lengths derived from an explicit ellipsoidal nonlocal averaging domain is still not commonly available.

To address the above limitations, the present study derives anisotropic characteristic lengths directly from ellipsoidal nonlocal strain averaging, thereby providing a geometric interpretation of the anisotropic length tensor. Based on this framework, a positive-definite second-order strain-gradient formulation is developed and incorporated into a unified C^1-continuous finite element framework employing cubic Hermite, Bogner–Fox–Schmit, and tensor-product tricubic Hermite elements for one, two, and three-dimensional analyses.

The main contributions of this work are:

- Ellipsoidal nonlocal averaging is used to derive direction-dependent characteristic lengths. The resulting tensor $L_{mn} = (AA^T)_{mn}$ represents second moments of the averaging domain and provides a geometric interpretation analogous to fabric tensors in homogenization literature [28–30].

- A positive-definite anisotropic second-order strain-gradient energy $W = 1/2C_{ijkl}\varepsilon_{ij}\varepsilon_{kl} + 1/20L_{mn}C_{ijkl}\varepsilon_{ij,m}\varepsilon_{kl,n}$ is formulated together with constitutive relations, governing equations and higher-order boundary conditions. After variation the $1/20$ factor becomes $1/10$ in the stress, which gives a positive stiffness correction in the wavenumber domain.

- A unified C^1-continuous finite element framework is developed for 1D, 2D and 3D. It uses cubic Hermite, Bogner–Fox–Schmit bicubic Hermite and tensor-product tricubic Hermite elements with 24 DOF per node in 3D, and diagonal scaling keeps the condition number $\kappa < 10^6$.

- Orientation-dependent tuning is governed by $L_{11}(\theta) = l_x^2\cos^2\theta + l_y^2\sin^2\theta$ and the off-diagonal term $L_{12} = (l_y^2 - l_x^2)\cos\theta\sin\theta$ that couples $B_{,x}^T C B_{,y}$, where $B_{,x} = \partial B/\partial x$ is the strain–displacement matrix differentiated with respect to x and C is the elasticity matrix (formally defined in Sec. 5). The response varies monotonically with orientation: the fundamental frequency f_1 (first natural frequency) decreases from 186.26 to 173.06 Hz and u/u_0 increases from 1.00 to 1.039, where u is the strain-gradient displacement and u_0 is the classical displacement. The normalized frequency $f_1/f_{iso} = 1.053 \to 0.979$, with f_{iso} the volume-equivalent isotropic reference frequency, shows no minimum at 45°. All cases remain above the classical bound $f_1 > 171.61$ Hz.

- A bivariate design-space map in $AR = l_x/l_y \in [1, 20]$ and $\theta \in [0°, 90°]$ is constructed together with the Directional Stiffening Index $DSI = (u_{ref} - u)/u_{ref}$. The index quantifies displacement reduction relative to the volume-equivalent isotropic reference $l_{iso} = (l_x l_y l_z)^{1/3}$. The map reveals three regions: stiffening-optimal at high AR and low θ, frequency-tuning at high AR near $\theta \approx 45°$, and insensitive when $AR \lesssim 2$ or $\theta \approx 90°$.

2. Formulation of the Problem

2.1. Nonlocal Strain Averaging

Let $\Omega \subset \mathbb{R}^3$ denote the material body and let $\mathbf{x} \in \Omega$ be an interior point such that the nonlocal neighbourhood introduced below remains contained in Ω. The nonlocal strain is introduced as a volume average of the classical strain over a finite

neighbourhood of $\mathbf{x}$, following the weakly nonlocal interpretation of Eringen-type continuum theories [16–18]:

$$\tilde{\varepsilon}_{mn}(\mathbf{x}) = \frac{1}{|\mathcal{V}|}\int_{\mathcal{V}} p(\boldsymbol{\xi})\,\varepsilon_{mn}(\mathbf{x}+\boldsymbol{\xi})\,dV_{\xi}. \tag{1}$$

Here ε_{mn} is the classical infinitesimal strain tensor, $\boldsymbol{\xi}$ is the relative position vector inside the averaging neighbourhood, and $\mathcal{V}$ is the nonlocal interaction domain. The kernel $p(\boldsymbol{\xi})$ is assumed to satisfy the normalization condition

$$\frac{1}{|\mathcal{V}|}\int_{\mathcal{V}} p(\boldsymbol{\xi})\,dV_{\xi} = 1. \tag{2}$$

In the present formulation, a uniform kernel is adopted, $p(\boldsymbol{\xi}) = 1$. This choice isolates the influence of the anisotropic averaging geometry and leads to closed-form moment relations.

The nonlocal neighbourhood is taken as an ellipsoid with semi-axes $l_1, l_2, l_3 > 0$ aligned with the Cartesian coordinate axes:

$$\mathcal{V} = \left\{\boldsymbol{\xi}\in\mathbb{R}^3 \;\middle|\; \frac{\xi_1^2}{l_1^2}+\frac{\xi_2^2}{l_2^2}+\frac{\xi_3^2}{l_3^2}\le 1\right\}, \qquad |\mathcal{V}| = \frac{4\pi}{3}\,l_1 l_2 l_3. \tag{3}$$

The assumption that $\mathcal{V}$ is fully contained in Ω neglects boundary-induced truncation of the nonlocal neighbourhood. The present derivation should therefore be interpreted as a bulk weakly nonlocal approximation. Near physical boundaries the averaging domain $\mathcal{V}(\mathbf{x})\cap\Omega$ is truncated. The first moment then becomes $\int \xi_i \neq 0$ and the second moment becomes position-dependent, $L_{mn}(\mathbf{x}) = 1/|\mathcal{V}\cap\Omega|\int_{\mathcal{V}\cap\Omega}\xi_m\xi_n dV$. This correction introduces a boundary layer in the characteristic-length tensor itself and an additional first-gradient term. In the weakly nonlocal regime $l_{\max} \ll L_{\varepsilon}$ and for interior points with $l_{\max} \ll \mathrm{dist}(\mathbf{x},\partial\Omega)$, the effect is second-order and is neglected. The natural double-traction condition $q_{ijm}\nu_m\nu_j = 0$ (with q_{ijm} the double stress defined in Eq. (17)) therefore remains first-order consistent. A full surface-corrected formulation with $L_{mn}(\mathbf{x})$ and Mindlin-Toupin surface divergence terms lies beyond the present phenomenological Aifantis-type model and is noted as future work for thin structures where $l_{\max} \sim h$, with h denoting the characteristic structural thickness (e.g., beam thickness, plate thickness or wall thickness). For such thin structures the required modifications are as follows. First, $L_{mn}(\mathbf{x})$ is computed analytically for the given geometry. Second, the energy becomes spatially varying, $W(\mathbf{x}) = 1/2C\varepsilon\varepsilon + 1/20L_{mn}(\mathbf{x})C\varepsilon_{,m}\varepsilon_{,n}$. Third, extra terms proportional to $L_{mn,j}\varepsilon_{,mn}$ and $L_{mn,jj}\varepsilon$ appear in the governing equation. Finally, when $h \sim l_{\max}$, i.e., when the structural thickness is comparable to the nonlocal length, the full integral model $\tilde{\varepsilon}(\mathbf{x}) = 1/|\mathcal{V}\cap\Omega|\int_{\mathcal{V}\cap\Omega}\varepsilon(\mathbf{x}+\xi)dV$ is used without Taylor expansion.

The ellipsoid is mapped from the unit ball by

$$\boldsymbol{\xi} = A\mathbf{u}, \qquad A = \mathrm{diag}(l_1,l_2,l_3), \qquad |\mathbf{u}|\le 1.$$

Consequently,

$$AA^T = \mathrm{diag}(l_1^2,l_2^2,l_3^2), \qquad (AA^T)_{ij} = l_i^2\delta_{ij} \quad \text{no sum on } i. \tag{4}$$

The tensor AA^T represents the directional second moment of the averaging geometry and acts as an anisotropic characteristic-length tensor. This interpretation is consistent with tensorial and ellipsoidal descriptions of internal length in anisotropic gradient elasticity [28–30]. The present development is restricted to an axis-aligned ellipsoidal neighbourhood, for which AA^T is diagonal. More generally, for a rotated ellipsoidal neighbourhood, the characteristic-length tensor becomes QAA^TQ^T, where Q is an orthogonal rotation tensor.

A caution is needed regarding the physical status of the ellipsoidal neighbourhood. The second-moment construction guarantees that an axis-aligned or rotated ellipsoid yields an anisotropic tensor L_{mn}, but that is a geometric statement about the averaging domain chosen here; it is not evidence that a given real microstructure possesses an anisotropic interaction domain. The features usually invoked in this connection — aligned fibres, elongated or drawn grains, oriented

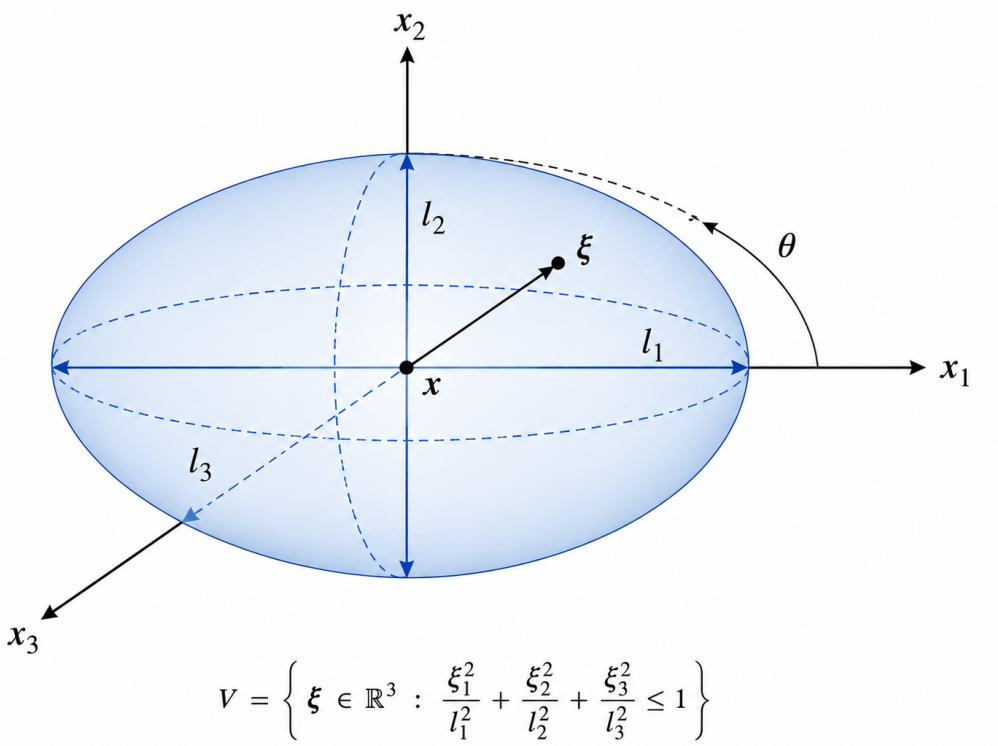


Figure 1: Ellipsoidal nonlocal averaging neighbourhood and directional characteristic lengths used in the anisotropic strain-gradient formulation.

lattice struts — are well documented to produce anisotropic macroscopic response, and anisotropic response is a weaker statement than anisotropic interaction. The ellipsoidal kernel adopted in Eq. (3) should therefore be read as a modelling ansatz: a parsimonious geometric parameterisation that admits direction-dependent interaction ranges and that collapses to the usual isotropic gradient model when $l_1 = l_2 = l_3$.

The ansatz is nevertheless testable, because an interaction range is an operationally well-defined quantity in its own right. In a weakly nonlocal description the neighbourhood over which the state at $\mathbf{x}$ is influenced by the state at $\mathbf{x} + \boldsymbol{\xi}$ plays the role of a correlation length of a local field measured between material points, so it can be estimated from microstructural images without reference to any stiffness measurement, and it can be resolved along and across a preferred material direction. Micro-computed tomography reconstructions of fibre paths in fibre-reinforced polymers have been used in exactly this way: microstructural descriptors are correlated as a function of separation distance and the distance over which that correlation decays defines a length scale of the microstructure, the three-dimensional reconstruction allowing the analysis to be performed along and across the fibres rather than on two-dimensional sections alone [56, 57]. In a related but distinct setting, the elasticity tensor of short fibre-reinforced composites has been shown to possess an anisotropic spatial correlation structure whose correlation functions differ between directions and reflect the overall transversely isotropic material behaviour [58]. These studies measure the spatial correlation of microstructural or material-property fields; they do not measure a strain-averaging kernel, and they do not establish the particular ellipsoidal form of $\mathcal{V}$ or the coefficient $1/10$ used here. What they do show is that direction-resolved correlation lengths are measurable properties of real heterogeneous media, so that an anisotropic L_{mn} is not purely a mathematical artefact of the averaging geometry. A direct experimental determination of the interaction-level anisotropy postulated in Eq. (3), as opposed to an inference from macroscopic response, is not attempted in the present work and remains essential future work; the measurement route is outlined as step 7 of the calibration protocol in Sec. 7.1.

Assuming that ε_{mn} is sufficiently smooth in a neighbourhood of $\mathbf{x}$, Taylor expansion gives

$$\varepsilon_{mn}(\mathbf{x}+\boldsymbol{\xi}) = \varepsilon_{mn}(\mathbf{x}) + \xi_i\,\varepsilon_{mn,i} + \frac{1}{2}\xi_i\xi_j\,\varepsilon_{mn,ij} + \frac{1}{6}\xi_i\xi_j\xi_k\,\varepsilon_{mn,ijk} + O(|\boldsymbol{\xi}|^4), \tag{5}$$

where a comma denotes spatial differentiation and repeated spatial indices are summed. Since the ellipsoidal domain is centrosymmetric and the adopted kernel is uniform, the first moment and all odd-order moments vanish after integration. The required moment identities are

$$\frac{1}{|\mathcal{V}|}\int_{\mathcal{V}} dV_\xi = 1, \qquad \frac{1}{|\mathcal{V}|}\int_{\mathcal{V}} \xi_i \, dV_\xi = 0, \qquad \frac{1}{|\mathcal{V}|}\int_{\mathcal{V}} \xi_i\xi_j \, dV_\xi = \frac{1}{5}(AA^T)_{ij}. \tag{6}$$

The last identity follows from the transformation to the unit ball, for which

$$\int_{|\mathbf{u}|\le 1} u_i u_j \, dV_u = \frac{4\pi}{15}\delta_{ij}.$$

Substitution of Eq. (5) into Eq. (1), together with the moment identities in Eq. (6), gives the second-order weakly nonlocal strain approximation

$$\tilde{\varepsilon}_{mn}(\mathbf{x}) = \varepsilon_{mn}(\mathbf{x}) + \frac{1}{10}(AA^T)_{ij}\frac{\partial^2 \varepsilon_{mn}}{\partial x_i \partial x_j} + O(l_{\max}^4 \nabla^4 \varepsilon_{mn}), \tag{7}$$

where

$$l_{\max} = \max\{l_1, l_2, l_3\}.$$

The coefficient $1/10$ is the product of the Taylor coefficient $1/2$ and the normalized second moment $1/5$ of the three-dimensional ellipsoidal averaging domain. The approximation is valid in the weakly nonlocal regime,

$$l_{\max} \ll L_\varepsilon,$$

where L_ε denotes the characteristic length over which the strain field varies appreciably. This Taylor-based reduction is the standard connection between integral nonlocal averaging and strain-gradient elasticity [19, 23].

Defining the anisotropic second-order differential operator

$$\mathcal{L}_\ell \equiv (AA^T)_{ij}\frac{\partial^2}{\partial x_i \partial x_j} = l_1^2\frac{\partial^2}{\partial x_1^2} + l_2^2\frac{\partial^2}{\partial x_2^2} + l_3^2\frac{\partial^2}{\partial x_3^2}, \tag{8}$$

Eq. (7) may be written compactly as

$$\tilde{\varepsilon}_{mn}(\mathbf{x}) = \left(1 + \frac{1}{10}\mathcal{L}_\ell\right)\varepsilon_{mn}(\mathbf{x}) + O(l_{\max}^4 \nabla^4 \varepsilon_{mn}). \tag{9}$$

The formulation above is a three-dimensional weakly nonlocal continuum approximation. Therefore, the one and two dimensional finite element models introduced later should be interpreted as reduced kinematic specializations of the same three-dimensional theory, rather than as independently averaged one or two dimensional nonlocal models. This distinction is important because the numerical coefficient in Eq. (7) is determined by the second moment of a three-dimensional ellipsoidal domain. For independently averaged domains the coefficients differ, where $\langle\cdot\rangle$ denotes the average over the domain, i.e., $\langle\xi_i^2\rangle = 1/|\mathcal{V}|\int_{\mathcal{V}}\xi_i^2 dV$. A 1D segment $[-l, l]$ has $|\mathcal{V}| = 2l$ and $\langle\xi^2\rangle = l^2/3$, which gives $1/6$ after the Taylor factor $1/2$. A 2D ellipse $\xi_1^2/l_1^2 + \xi_2^2/l_2^2 \le 1$ has $|\mathcal{V}| = \pi l_1 l_2$ with $\langle\xi_i^2\rangle = l_i^2/4$, leading to $1/8$. The 3D ellipsoid gives $|\mathcal{V}| = 4\pi/3 l_1 l_2 l_3$ and $\langle\xi_i^2\rangle = l_i^2/5$, hence $1/10$ (see Appendix A). We retain $1/10$ in 1D and 2D because the physical neighbourhood remains a 3D ellipsoid, while the 1D and 2D models are reduced kinematics (uniaxial and plane-strain) of the same 3D continuum. This choice keeps the models comparable. At $l/L = 0.20$, where L denotes the relevant structural length (bar length, beam length or plate dimension), the 1D stiffening ratio is 0.936 and the 2D ratio is 0.94. The 3D ratio is 0.959, similar to the lower-dimensional values. The coefficient $1/10$ has been used consistently across all numerical results in the manuscript (Figs. 1-11). For a truly 1D chain or 2D membrane $1/6$ and $1/8$ would be appropriate. See

Appendix A for the full derivation. The energy coefficient $1/20$ yields $1/10$ in the stress after variation by symmetry (see Sec. 2.3).

2.2. Kinematic Relations

The displacement field is denoted by $\mathbf{u}(\mathbf{x})$, with components u_i. Under the small-strain assumption, the infinitesimal strain tensor is

$$\varepsilon_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i}\right), \tag{10}$$

where a comma denotes partial differentiation with respect to the corresponding spatial coordinate. The volumetric strain, or dilatation, is given by

$$\varepsilon_{kk} = u_{k,k} = \nabla \cdot \mathbf{u}. \tag{11}$$

2.3. Strain Energy Density and Constitutive Relations

Motivated by the anisotropic differential operator obtained from the ellipsoidal nonlocal averaging procedure, a strain-gradient energy density is introduced in the form

$$W = \frac{1}{2}C_{ijkl}\varepsilon_{ij}\varepsilon_{kl} + \frac{1}{20}L_{mn}C_{ijkl}\varepsilon_{ij,m}\varepsilon_{kl,n}, \qquad L_{mn} = (AA^T)_{mn}. \tag{12}$$

Here L_{mn} is the anisotropic characteristic-length tensor obtained from the second moment of the ellipsoidal averaging domain. The coefficient $1/20$ is selected so that the variationally derived stress correction contains the same second-moment coefficient $1/10$ as the weakly nonlocal strain expansion, while preserving a positive gradient contribution to the stored energy. The form of Eq. (12) is consistent with simplified strain-gradient elasticity models and their anisotropic extensions [28, 29]. Eq. (7) and Eq. (12) arise from distinct kinematic and constitutive steps. Eq. (7) $\tilde{\varepsilon} = (1 + 1/10\mathcal{L})\varepsilon$ is a kinematic result. It follows from Taylor expansion of the nonlocal average with a uniform kernel over the ellipsoid. The coefficient $1/10$ is the product of the Taylor factor $1/2$ and the normalized second moment $1/5$. Eq. (12) $W = 1/2C\varepsilon\varepsilon + 1/20L_{mn}C\varepsilon_{,m}\varepsilon_{,n}$ is a separate constitutive choice. It is motivated by the operator L_{mn} but is chosen to ensure positive-definiteness with $\mu > 0$, $3\lambda + 2\mu > 0$ and $l_i > 0$ giving L_{mn} positive-definite, and to give a well-defined C^1 variational formulation. It is not uniquely derived from averaging. Variation gives $\delta W = C_{ijkl}\varepsilon_{kl}\delta\varepsilon_{ij} + 1/20L_{mn}C_{ijkl}(\varepsilon_{ij,m}\delta\varepsilon_{kl,n} + \delta\varepsilon_{ij,m}\varepsilon_{kl,n})$. This becomes $C\varepsilon\delta\varepsilon + 1/10L_{mn}C\varepsilon_{,n}\delta\varepsilon_{,m}$, or $\tau_{ij}\delta\varepsilon_{ij} + q_{ijm}\delta\varepsilon_{ij,m}$ with $q_{ijm} = 1/10L_{mn}C_{ijkl}\varepsilon_{kl,n}$, where τ_{ij} and q_{ijm} are the energetic stress and double stress formally defined in Eqs. (16) and (17). The form uses symmetry $C_{ijkl} = C_{klij}$ and $L_{mn} = L_{nm}$. Integration by parts gives $\int q_{ijm}\delta\varepsilon_{ij,m} = -\int q_{ijm,m}\delta\varepsilon_{ij}$ plus a boundary term. Hence the effective stress is $\sigma_{ij} = \tau_{ij} - q_{ijm,m} = C\varepsilon - 1/10L_{mn}(C\varepsilon)_{,mn} = (1 - 1/10\mathcal{L})\sigma^{cl}_{ij}$, where σ^{cl}_{ij} is the classical elastic stress defined in Eq. (14). Thus $1/20$ in the energy yields $1/10$ in the stress and in the double stress. If $1/10$ were used in the energy, the stress correction would be $1/5$, a double-counting. This separation between kinematic averaging and constitutive regularization clarifies their distinct roles. The local C_{ijkl} is taken isotropic to isolate nonlocal anisotropy coming from the ellipsoidal geometry. Physically this corresponds to materials where the matrix is isotropic but the interaction range is direction-dependent due to elongated features such as fibres, drawn grains and lattice struts. L_{mn} is analogous to the fabric tensor in homogenization [28–30]. For materials with additional local anisotropy both sources should be combined as $W = 1/2C^{aniso}\varepsilon\varepsilon + 1/20L_{mn}C^{aniso}\varepsilon_{,m}\varepsilon_{,n}$.

For the isotropic elastic solid considered here, the fourth-order elasticity tensor is

$$C_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right), \tag{13}$$

where λ and μ are the Lamé constants. The classical elastic stress corresponding to the local part of the energy is

$$\sigma^{\mathrm{cl}}_{ij} = C_{ijkl}\varepsilon_{kl} = \lambda\delta_{ij}\varepsilon_{kk} + 2\mu\varepsilon_{ij}. \tag{14}$$

For $\mu > 0$, $3\lambda + 2\mu > 0$, and positive characteristic lengths $l_i > 0$, the tensor L_{mn} is positive definite and the gradient contribution in Eq. (12) is non-negative. Thus, the adopted energy density is positive definite for non-rigid admissible strain fields.

The total potential energy is written as

$$\Pi = \int_{\Omega} W \, d\Omega - \int_{\Omega} P_i u_i \, d\Omega - \int_{\Gamma_t} \bar{t}_i u_i \, d\Gamma, \tag{15}$$

where P_i is the body-force density and $\bar{t}_i$ is the prescribed classical traction on Γ_t. Possible higher-order boundary terms associated with the strain-gradient contribution are not prescribed in Eq. (15); the corresponding natural boundary conditions arise from the variational statement and are discussed separately in the boundary-condition section.

The energetic stress and double stress are defined by

$$\tau_{ij} = \frac{\partial W}{\partial \varepsilon_{ij}} = C_{ijkl}\varepsilon_{kl}, \tag{16}$$

and

$$q_{ijm} = \frac{\partial W}{\partial \varepsilon_{ij,m}} = \frac{1}{10} L_{mn} C_{ijkl} \varepsilon_{kl,n}. \tag{17}$$

The factor $1/10$ in Eq. (17) results from the symmetry of C_{ijkl} and L_{mn} in the variation of the quadratic gradient term. For a homogeneous material with constant C_{ijkl} and constant characteristic-length tensor L_{mn}, the effective stress entering the strong equilibrium equation is

$$\begin{aligned} \sigma_{ij} &= \tau_{ij} - q_{ijm,m} \\ &= C_{ijkl}\varepsilon_{kl} - \frac{1}{10} L_{mn} \frac{\partial^2}{\partial x_m \partial x_n} \left(C_{ijkl}\varepsilon_{kl} \right) \\ &= \left(1 - \frac{1}{10} \mathcal{L}_\ell \right) \sigma_{ij}^{\mathrm{cl}}, \end{aligned} \tag{18}$$

where the anisotropic operator $\mathcal{L}_\ell$ is defined in Eq. (8). Equivalently,

$$\sigma_{ij} = \left[1 - \frac{1}{10} \left(l_1^2 \frac{\partial^2}{\partial x_1^2} + l_2^2 \frac{\partial^2}{\partial x_2^2} + l_3^2 \frac{\partial^2}{\partial x_3^2} \right) \right] \left(\lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij} \right). \tag{19}$$

The signs in the nonlocal strain approximation and in the effective stress relation have different origins. The averaged strain obtained from Taylor expansion satisfies

$$\tilde{\varepsilon}_{ij} = \left(1 + \frac{1}{10} \mathcal{L}_\ell \right) \varepsilon_{ij} + O(l_{\max}^4 \nabla^4 \varepsilon_{ij}), \tag{20}$$

whereas the stress relation in Eq. (18) follows from the variation of the positive-definite strain-gradient energy. For a plane wave $\exp(i\mathbf{k} \cdot \mathbf{x})$,

$$\mathcal{L}_\ell \to -L_{mn} k_m k_n,$$

so that

$$1 - \frac{1}{10} \mathcal{L}_\ell \to 1 + \frac{1}{10} L_{mn} k_m k_n.$$

Hence, the negative sign in the physical-space stress operator produces a positive stiffness correction in the wavenumber domain.

2.4. Governing Equation

The equation of motion follows from the local balance of linear momentum written in terms of the effective stress defined in Eq. (18):

$$\sigma_{ij,j} + P_i = \rho\, u_{i,tt}, \tag{21}$$

where P_i is the body force per unit volume and ρ is the mass density. For a homogeneous material with constant Lamé parameters and constant characteristic-length tensor L_{mn}, the operator $\mathcal{L}_\ell$ has constant coefficients and commutes with spatial differentiation. Substituting Eq. (19) into Eq. (21) therefore gives

$$\left(1 - \frac{1}{10}\mathcal{L}_\ell\right)\left[(\lambda+\mu)u_{j,ij} + \mu u_{i,jj}\right] + P_i = \rho\, u_{i,tt}. \tag{22}$$

The term in square brackets is the classical Navier operator,

$$\sigma^{\mathrm{cl}}_{ij,j} = (\lambda+\mu)u_{j,ij} + \mu u_{i,jj}. \tag{23}$$

Equation (22) is therefore a fourth-order anisotropic strain-gradient extension of the classical elastodynamic equation. Such higher-order governing equations require additional boundary conditions beyond the classical displacement or traction conditions [14, 15, 23]. In the static case, the inertia term on the right-hand side is omitted.

For reference, consider the isotropic spherical case $l_1 = l_2 = l_3 = l$, for which $\mathcal{L}_\ell = l^2\nabla^2$. In the absence of body forces, let

$$\mathbf{u} = \mathbf{U}\exp\left[i(\mathbf{k}\cdot\mathbf{x} - \omega t)\right], \tag{24}$$

where $\mathbf{U}$ is the constant (complex) amplitude vector, $\mathbf{k}$ is the wave vector, $k = |\mathbf{k}|$, and ω is the circular frequency. Substitution into Eq. (22) yields the longitudinal and transverse dispersion relations

$$\text{P-wave:} \qquad \omega^2 = c_P^2 k^2\left(1 + \frac{l^2k^2}{10}\right), \tag{25}$$

$$\text{S-wave:} \qquad \omega^2 = c_S^2 k^2\left(1 + \frac{l^2k^2}{10}\right), \tag{26}$$

with

$$c_P = \sqrt{\frac{\lambda+2\mu}{\rho}}, \qquad c_S = \sqrt{\frac{\mu}{\rho}}. \tag{27}$$

The positive correction factor in Eqs. (25)–(26) indicates short-wavelength stiffening within the range of validity of the second-order weakly nonlocal approximation, i.e. for $lk \ll 1$ [23, 24]. Dispersion relations serve as an analytical stability check. The positive factor $1 + l^2k^2/10$ for all k confirms positive-definiteness with no imaginary frequencies. The change from $(1 - l^2/10\nabla^2)$ in physical space to $1 + l^2k^2/10$ in the wavenumber domain explains the sign remark and indicates short-wavelength stiffening within $lk \ll 1$. This supports the $l/L \leq 0.10$ validity range. The present work focuses on static and free-vibration analysis.

3. Particular Cases

Case 1: Spheroidal neighbourhood

When $l_1 = l_2 = l$ and l_3 is distinct, the ellipsoidal neighbourhood has axial symmetry about the x_3-axis. The anisotropic operator reduces to

$$\mathcal{L}_\ell = l^2 \frac{\partial^2}{\partial x_1^2} + l^2 \frac{\partial^2}{\partial x_2^2} + l_3^2 \frac{\partial^2}{\partial x_3^2}. \tag{28}$$

Accordingly, the effective stress becomes

$$\sigma_{ij} = \left[1 - \frac{1}{10}\left(l^2 \frac{\partial^2}{\partial x_1^2} + l^2 \frac{\partial^2}{\partial x_2^2} + l_3^2 \frac{\partial^2}{\partial x_3^2}\right)\right]\left(\lambda \delta_{ij}\varepsilon_{kk} + 2\mu\varepsilon_{ij}\right). \tag{29}$$

The case $l_3 > l$ corresponds to a prolate spheroidal neighbourhood, whereas $l_3 < l$ corresponds to an oblate spheroidal neighbourhood.

Case 2: Spherical neighbourhood

When $l_1 = l_2 = l_3 = l$, the averaging neighbourhood is spherical and

$$\mathcal{L}_\ell = l^2 \nabla^2. \tag{30}$$

The constitutive relation then reduces to

$$\sigma_{ij} = \left(1 - \frac{l^2}{10}\nabla^2\right)\left(\lambda \delta_{ij}\varepsilon_{kk} + 2\mu\varepsilon_{ij}\right). \tag{31}$$

Comparison with integral nonlocal, homogenized lattice and alternative gradient formulations is as follows. The integral model is $\tilde{\sigma}_{ij}(\mathbf{x}) = \int K(|\mathbf{x} - \mathbf{x}'|)\sigma_{ij}^{cl}(\mathbf{x}')d\Omega'$ with a Gaussian kernel, where $\mathbf{x}'$ is the source (reference) point. Taylor expansion gives $\tilde{\sigma} = \sigma^{cl} + l^2/10\nabla^2\sigma^{cl} + O(l^4)$ for a spherical uniform kernel over the ellipsoid. Hence the differential form $(1 - l^2/10\nabla^2)\sigma = \sigma^{cl}$ is a second-order approximation with error $O(l^4\nabla^4\sigma)$ [18, 23]. An anisotropic ellipsoidal kernel gives an anisotropic Laplacian $L_{mn}\partial^2$. Second-order homogenization of lattices [32] shows that the gradient continuum reproduces size effects missed by the classical model, with $L_{mn} \propto \langle \xi_m \xi_n \rangle_{RVE}$. A full comparison would require an RVE of an octet-truss and similar architectures, computing C_{ijkl} and L_{mn} via Hill-Mandel. This is noted as future work requiring a separate solver with real RVE simulations. Compared with Mindlin Form II with 5 lengths and Form III, our simplified Aifantis-type model uses 3 lengths l_i from geometry. It remains positive-definite and is computationally tractable with C^1 FEM, whereas full Mindlin requires mixed FEM. The trade-off is less generality but more robustness. The present differential model is thus a weak nonlocal approximation to the integral model, valid for $l_{\max} \ll L_\varepsilon$ with error $O(l^4)$. This is the isotropic Aifantis-type strain-gradient form [19, 21, 23] with the effective gradient parameter

$$\ell_{\text{eff}}^2 = \frac{l^2}{10}. \tag{32}$$

4. Boundary Conditions

The governing equation (22) contains fourth-order spatial derivatives because the anisotropic operator $\mathcal{L}_\ell$ acts on the classical Navier operator. Therefore, in addition to the classical displacement or traction conditions, one higher-order boundary condition is required. Let $\boldsymbol{\nu}$ be the outward unit normal to $\partial\Omega$, and let Γ_u and Γ_t denote the displacement and traction parts of the classical boundary, respectively.

4.1. Classical Boundary Conditions

On Γ_u, the displacement is prescribed as

$$u_i = \bar{u}_i \qquad \text{on } \Gamma_u, \tag{33}$$

where the overbar denotes a prescribed (given) value. On Γ_t, the effective traction is taken as

$$\sigma_{ij}\nu_j = \bar{t}_i \qquad \text{on } \Gamma_t, \tag{34}$$

where σ_{ij} is the effective stress defined in Eq. (18). For a traction-free surface,

$$\sigma_{ij}\nu_j = 0. \tag{35}$$

4.2. Higher-Order Boundary Condition

For the homogeneous material considered here, the strain-gradient part of the energy introduces the higher-order stress

$$q_{ijm} = \frac{\partial W}{\partial \varepsilon_{ij,m}} = \frac{1}{10} L_{mn} C_{ijkl} \varepsilon_{kl,n} = \frac{1}{10} L_{mn} \sigma^{\mathrm{cl}}_{ij,n}, \qquad L_{mn} = (AA^T)_{mn}. \tag{36}$$

The tensor q_{ijm} is work-conjugate to the strain gradient $\varepsilon_{ij,m}$. The natural higher-order condition is written in terms of the normal double traction as

$$q_{ijm}\nu_m\nu_j = 0 \qquad \text{on } \Gamma_q, \tag{37}$$

where $\Gamma_q \subseteq \partial\Omega$ is the portion of the boundary on which the higher-order natural condition is imposed. On the complementary higher-order essential boundary Γ_g, one may prescribe the normal derivative of displacement,

$$u_{i,\nu} = \nabla u_i \cdot \boldsymbol{\nu} = \bar{g}_i \qquad \text{on } \Gamma_g, \tag{38}$$

where $\bar{g}_i$ denotes the prescribed value of the normal derivative of displacement. In the present computations, the natural higher-order condition is used on free boundaries unless stated otherwise.

Using Eq. (36), Eq. (37) becomes

$$L_{mn} \frac{\partial \sigma^{\mathrm{cl}}_{ij}}{\partial x_n} \nu_m \nu_j = 0 \qquad \text{on } \Gamma_q. \tag{39}$$

The factor $1/10$ has been omitted since the boundary condition is homogeneous and multiplication by a nonzero constant does not alter the resulting constraint. This condition represents the reduced higher-order natural boundary condition adopted in the present formulation. In accordance with the underlying Aifantis-type strain-gradient framework [19, 21, 23], higher-order boundary effects are expressed through normal derivatives and normal double tractions, which are employed consistently throughout the subsequent finite element formulation. Higher-order boundary conditions comprise essential and natural types. Classical conditions are $u_i = \bar{u}_i$ on Γ_u and $\sigma_{ij}\nu_j = \bar{t}_i$ on Γ_t with $\sigma_{ij} = (1 - 1/10\mathcal{L})\sigma^{cl}_{ij}$. Higher-order conditions are of two types. (i) Essential: $u_{i,\nu} = \nabla u_i \cdot \boldsymbol{\nu} = \bar{g}_i$ on Γ_g, enforced through Hermite derivative DOFs $(u_{i,x}, u_{i,y}, u_{i,z})$. For a clamped-slope bar $\theta(0) = 0$ is imposed by setting the nodal derivative DOF to zero. (ii) Natural: $q_{ijm}\nu_m\nu_j = 0$ on Γ_q with $q_{ijm} = 1/10 L_{mn}\sigma^{cl}_{ij,n}$. This follows from the boundary term $\int_{\partial\Omega} q_{ijm}\nu_m \delta\varepsilon_{ij} d\Gamma$ after integration by parts. On free boundaries where derivative DOFs are left free, the weak form enforces this natural condition automatically. For a homogeneous material with constant L_{mn} the operator commutes. The strong form $(1 - 1/10\mathcal{L})[(\lambda + \mu)u_{j,ij} + \mu u_{i,jj}] + P_i = \rho\ddot{u}_i$ is then fourth-order and needs one extra boundary condition per side, which is provided. For the spherical case on a flat surface x_3 =const with $\boldsymbol{\nu} = (0, 0, 1)$, the conditions become $\sigma_{i3} = 0$ and $\partial\sigma^{cl}_{i3}/\partial x_3 = 0$ (Eq.26-27). This is a reduced Aifantis-type condition, not the full Mindlin-Toupin form with surface

divergence, noted as a limitation. In the FEM implementation essential higher-order conditions are imposed by constraining derivative DOFs in K_{ff}, F_f (where K_{ff} and F_f denote the stiffness matrix and force vector after imposing essential boundary conditions, formally defined in Sec. 5.5) before scaling. Natural conditions are satisfied variationally. In full Mindlin-Toupin second-gradient theory the variational principle yields surface tractions t_i, double tractions $r_i = q_{ijm}\nu_m\nu_j$ and line forces $e_i = [\![q_{ijm}\nu_m\mu_j]\!]$ along edges where the normal ν is discontinuous, with μ the edge co-normal and $[\![\cdot]\!]$ the jump. For rectangular and brick domains with 90° corners the simplified Aifantis-type model adopts reduced conditions $\sigma_{ij}\nu_j = \bar{t}_i$ and $q_{ijm}\nu_m\nu_j = 0$. Surface divergence and edge terms are neglected, consistent with the phenomenological energy. Hermite derivative DOFs are shared across elements including corners, so the weak form satisfies natural corner conditions variationally. Reaction error $< 10^{-11}$ confirms this. For general polyhedral domains explicit edge conditions are required, noted as a limitation. For a rotated L_{mn} the double traction $q_{ijm}\nu_m = 1/10 L_{mn}\sigma^{cl}_{ij,n}\nu_m$ contains off-diagonal contributions. For a clamped face $x = 0$ with $\nu = (1, 0, 0)$, the rotated tensor $L_{rot} = R^T diag(l_x^2, l_y^2, l_z^2)R$ gives $L_{11} = l_x^2c^2 + l_y^2s^2$ and $L_{12} = (l_y^2 - l_x^2)cs$. Hence $r_{ij} = 1/10[L_{11}\sigma^{cl}_{ij,1} + L_{12}\sigma^{cl}_{ij,2}]$. The off-diagonal term couples the transverse gradient $\sigma^{cl}_{ij,2}$ into the double traction even though ν is along x. At $\theta = 45°$ the coupling reaches its maximum magnitude with $L_{12} \approx -0.0198$ for $l_x = 0.20$ and $l_y = 0.02$. This causes energy redistribution, but the frequency still decreases monotonically with no minimum at 45°. Tangential components are work-conjugate to $\delta\varepsilon_{ij}$. The natural condition $q_{ijm}\nu_m\nu_j = 0$ retains the normal part, while the tangential part would contribute to edge forces in the full theory.

For a spherical neighbourhood, $L_{mn} = l^2\delta_{mn}$. On a flat traction-free surface $x_3 = \text{const}$ with $\boldsymbol{\nu} = (0, 0, 1)$, the classical and higher-order natural conditions reduce to

$$\sigma_{i3} = 0, \qquad i = 1, 2, 3, \tag{40}$$

$$\frac{\partial \sigma^{\mathrm{cl}}_{i3}}{\partial x_3} = 0, \qquad i = 1, 2, 3. \tag{41}$$

The first condition represents the effective traction-free boundary, while the second condition is the corresponding normal double-traction condition.

5. Finite Element Formulation

This section presents the displacement-based finite element formulation corresponding to the strain-gradient model derived above. Since the governing equation contains fourth-order spatial derivatives, a direct Galerkin discretization requires C^1-continuous basis functions. Cubic Hermite elements are therefore used in one dimension, Bogner–Fox–Schmit bicubic Hermite rectangular elements in two dimensions, and tensor-product tricubic Hermite brick elements in three dimensions [50–53]. The present implementation uses tensor-product rectangular and brick elements so that Hermite derivative degrees of freedom are shared consistently across element interfaces.

5.1. Weak Form

The strain-gradient energy density is

$$W = \frac{1}{2}C_{ijkl}\varepsilon_{ij}\varepsilon_{kl} + \frac{1}{20}L_{mn}C_{ijkl}\varepsilon_{ij,m}\varepsilon_{kl,n}, \qquad L_{mn} = (AA^T)_{mn}. \tag{42}$$

For an axis-aligned ellipsoidal neighbourhood,

$$L_{mn} = l_m^2\delta_{mn} \qquad \text{no sum on } m. \tag{43}$$

This diagonal form follows directly from the second moment $1/|\mathcal{V}| \int_{\mathcal{V}} \xi_m\xi_n dV = (1/5)(AA^T)_{mn}$ of the axis-aligned ellipsoid and is not an arbitrary choice; off-diagonal terms vanish because the ellipsoid axes are aligned with the coordinate axes.

Let $\mathbf{u}$ be the displacement field and let $\mathbf{v}$ be an admissible virtual displacement field vanishing on the essential boundary. The weak form follows from the first variation of the total potential energy, supplemented by the inertial virtual work. Let $\mathcal{U}$ denote the space of admissible displacement fields satisfying the essential boundary conditions on Γ_u (and Γ_g when prescribed), and let $\mathcal{V}_0$ denote the space of test functions vanishing on the essential boundary. The problem is to find $\mathbf{u} \in \mathcal{U}$ such that, for all $\mathbf{v} \in \mathcal{V}_0$,

$$\int_\Omega \varepsilon_{ij}(\mathbf{v}) C_{ijkl} \varepsilon_{kl}(\mathbf{u})\, d\Omega + \frac{1}{10} \int_\Omega L_{mn} \varepsilon_{ij,m}(\mathbf{v}) C_{ijkl} \varepsilon_{kl,n}(\mathbf{u})\, d\Omega + \int_\Omega \rho\, v_i \ddot{u}_i\, d\Omega = \int_\Omega v_i P_i\, d\Omega + \int_{\Gamma_t} v_i \bar{t}_i\, d\Gamma. \tag{44}$$

For static problems, the inertial term is omitted. The higher-order natural boundary terms are those associated with the boundary conditions introduced in Section 4.

After spatial discretization, the semi-discrete finite element equations take the standard form

$$\mathbf{M}\ddot{\mathbf{d}} + \mathbf{K}\mathbf{d} = \mathbf{F}, \tag{45}$$

where $\mathbf{d}$ is the global degree-of-freedom vector, $\mathbf{F}$ is the global load vector, $\mathbf{M}$ is the mass matrix, and

$$\mathbf{K} = \mathbf{K}_{\mathrm{cl}} + \mathbf{K}_{\mathrm{g}}. \tag{46}$$

The classical stiffness matrix is

$$\mathbf{K}_{\mathrm{cl}} = \int_\Omega \mathbf{B}^T \mathbf{C} \mathbf{B}\, d\Omega, \tag{47}$$

where $\mathbf{B}$ is the classical strain-displacement matrix and $\mathbf{C}$ is the elasticity matrix in Voigt notation consistent with the engineering strain vector. The gradient stiffness matrix is

$$\mathbf{K}_{\mathrm{g}} = \frac{1}{10} \int_\Omega \sum_{\alpha=1}^{d} \sum_{\beta=1}^{d} L_{\alpha\beta} \left(\frac{\partial \mathbf{B}}{\partial x_\alpha} \right)^T \mathbf{C} \left(\frac{\partial \mathbf{B}}{\partial x_\beta} \right) d\Omega, \tag{48}$$

where d is the spatial dimension of the reduced kinematic model. For an axis-aligned ellipsoidal neighbourhood, Eq. (48) reduces to

$$\mathbf{K}_{\mathrm{g}} = \frac{1}{10} \int_\Omega \sum_{\alpha=1}^{d} l_\alpha^2 \left(\frac{\partial \mathbf{B}}{\partial x_\alpha} \right)^T \mathbf{C} \left(\frac{\partial \mathbf{B}}{\partial x_\alpha} \right) d\Omega. \tag{49}$$

For the spherical characteristic-length case, $l_1 = l_2 = l_3 = l$, this becomes

$$\mathbf{K}_{\mathrm{g}} = \frac{l^2}{10} \int_\Omega \sum_{\alpha=1}^{d} \left(\frac{\partial \mathbf{B}}{\partial x_\alpha} \right)^T \mathbf{C} \left(\frac{\partial \mathbf{B}}{\partial x_\alpha} \right) d\Omega. \tag{50}$$

The consistent mass matrix is

$$\mathbf{M} = \int_\Omega \rho\, \mathbf{N}^T \mathbf{N}\, d\Omega, \tag{51}$$

where $\mathbf{N}$ is the displacement interpolation matrix. For free-vibration analysis, Eq. (45) gives the generalized eigenvalue problem

$$\mathbf{K}\boldsymbol{\phi} = \omega^2 \mathbf{M}\boldsymbol{\phi}, \tag{52}$$

where $\boldsymbol{\phi}$ is the eigenvector (natural mode shape) and ω is the natural circular frequency.

5.2. One-Dimensional Hermite Element

For the one-dimensional bar, the displacement field is $u(x)$ and

$$\varepsilon = u', \qquad \varepsilon_{,x} = u''. \tag{53}$$

The strain energy reduces to

$$U = \frac{1}{2}\int_0^L EA_c\left[(u')^2 + a^2(u'')^2\right]dx, \qquad a^2 = \frac{l^2}{10}, \tag{54}$$

where E is Young's modulus and A_c is the cross-sectional area.

A two-node cubic Hermite element is used. Each node contains the displacement and its first derivative,

$$\mathbf{d}_i = \begin{bmatrix} u_i & \theta_i \end{bmatrix}^T, \qquad \theta_i = \left.\frac{du}{dx}\right|_{x=x_i}. \tag{55}$$

For an element of length L_e,

$$u^e(x) = \mathbf{N}\mathbf{d}^e, \qquad \mathbf{d}^e = \begin{bmatrix} u_1 & \theta_1 & u_2 & \theta_2 \end{bmatrix}^T, \tag{56}$$

where $\mathbf{N} = [N_1\ N_2\ N_3\ N_4]$ is the Hermite shape-function matrix containing the cubic shape functions defined below.

Using the local natural coordinate $s = x/L_e$, $0 \le s \le 1$ (with $s = 0$ at first node and $s = 1$ at second node), the Hermite shape functions are

$$N_1 = 1 - 3s^2 + 2s^3, \tag{57}$$

$$N_2 = L_e(s - 2s^2 + s^3), \tag{58}$$

$$N_3 = 3s^2 - 2s^3, \tag{59}$$

$$N_4 = L_e(-s^2 + s^3). \tag{60}$$

Defining

$$\mathbf{B}_1 = \frac{d\mathbf{N}}{dx}, \qquad \mathbf{B}_2 = \frac{d^2\mathbf{N}}{dx^2}, \tag{61}$$

the element stiffness matrix is

$$\mathbf{K}^e = \int_{x_e} EA_c\left(\mathbf{B}_1^T\mathbf{B}_1 + a^2\mathbf{B}_2^T\mathbf{B}_2\right)dx. \tag{62}$$

The consistent mass matrix is

$$\mathbf{M}^e = \int_{x_e} \rho A_c\,\mathbf{N}^T\mathbf{N}\,dx. \tag{63}$$

5.3. Two-Dimensional Bogner–Fox–Schmit Element

For the two-dimensional plane-strain formulation, the displacement field is

$$\mathbf{u} = \begin{bmatrix} u & v \end{bmatrix}^T, \tag{64}$$

and the engineering strain vector is

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} u_{,x} \\ v_{,y} \\ u_{,y} + v_{,x} \end{bmatrix}. \tag{65}$$

The matrix $\mathbf{C}$ in this subsection denotes the corresponding plane-strain elasticity matrix written in Voigt form consistent with the engineering shear strain convention.

A Bogner–Fox–Schmit bicubic Hermite rectangular element is used for each displacement component. For a scalar field w, the nodal degrees of freedom are

$$w, \qquad w_{,x}, \qquad w_{,y}, \qquad w_{,xy}. \tag{66}$$

Thus, each node has eight degrees of freedom in two-dimensional elasticity:

$$\mathbf{d}_i = \begin{bmatrix} u_i & u_{i,x} & u_{i,y} & u_{i,xy} & v_i & v_{i,x} & v_{i,y} & v_{i,xy} \end{bmatrix}^T. \tag{67}$$

For the spherical characteristic-length case, the element stiffness matrix is

$$\mathbf{K}^e = \int_{\Omega_e} \left[\mathbf{B}^T \mathbf{C} \mathbf{B} + a^2 \left(\mathbf{B}_{,x}^T \mathbf{C} \mathbf{B}_{,x} + \mathbf{B}_{,y}^T \mathbf{C} \mathbf{B}_{,y} \right) \right] d\Omega, \qquad a^2 = \frac{l^2}{10}. \tag{68}$$

For an axis-aligned ellipsoidal model, the stiffness matrix becomes

$$\mathbf{K}^e = \int_{\Omega_e} \left[\mathbf{B}^T \mathbf{C} \mathbf{B} + \frac{l_1^2}{10} \mathbf{B}_{,x}^T \mathbf{C} \mathbf{B}_{,x} + \frac{l_2^2}{10} \mathbf{B}_{,y}^T \mathbf{C} \mathbf{B}_{,y} \right] d\Omega. \tag{69}$$

If a rotated characteristic-length tensor is used, the cross term $L_{12} \mathbf{B}_{,x}^T \mathbf{C} \mathbf{B}_{,y}$ and its symmetric counterpart follow directly from Eq. (48).

The consistent mass matrix is

$$\mathbf{M}^e = \int_{\Omega_e} \rho \, \mathbf{N}_v^T \mathbf{N}_v \, d\Omega, \tag{70}$$

where $\mathbf{N}_v$ is the vector displacement interpolation matrix.

5.4. Three-Dimensional Tricubic Hermite Brick Element

For the three-dimensional formulation, the displacement vector is

$$\mathbf{u} = \begin{bmatrix} u & v & w \end{bmatrix}^T, \tag{71}$$

and the engineering strain vector is

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} u_{,x} \\ v_{,y} \\ w_{,z} \\ v_{,z} + w_{,y} \\ u_{,z} + w_{,x} \\ u_{,y} + v_{,x} \end{bmatrix}. \tag{72}$$

Here also, $\mathbf{C}$ is understood in Voigt form consistent with the engineering shear strain convention.

A tensor-product tricubic Hermite brick element is used. For each scalar field q, the nodal degrees of freedom are

$$q, \quad q_{,x}, \quad q_{,y}, \quad q_{,z}, \quad q_{,xy}, \quad q_{,xz}, \quad q_{,yz}, \quad q_{,xyz}. \tag{73}$$

Thus, for the three displacement components, each node contains 24 degrees of freedom:

$$\begin{aligned}\mathbf{d}_i = [&u_i, u_{i,x}, u_{i,y}, u_{i,z}, u_{i,xy}, u_{i,xz}, u_{i,yz}, u_{i,xyz},\\ &v_i, v_{i,x}, v_{i,y}, v_{i,z}, v_{i,xy}, v_{i,xz}, v_{i,yz}, v_{i,xyz},\\ &w_i, w_{i,x}, w_{i,y}, w_{i,z}, w_{i,xy}, w_{i,xz}, w_{i,yz}, w_{i,xyz}]^T.\end{aligned} \tag{74}$$

For the spherical characteristic-length case, the element stiffness matrix is

$$\mathbf{K}^e = \int_{\Omega_e} \left[\mathbf{B}^T\mathbf{C}\mathbf{B} + a^2\left(\mathbf{B}_{,x}^T\mathbf{C}\mathbf{B}_{,x} + \mathbf{B}_{,y}^T\mathbf{C}\mathbf{B}_{,y} + \mathbf{B}_{,z}^T\mathbf{C}\mathbf{B}_{,z}\right)\right] d\Omega, \qquad a^2 = \frac{l^2}{10}. \tag{75}$$

For the axis-aligned ellipsoidal model with distinct characteristic lengths, the stiffness matrix is

$$\mathbf{K}^e = \int_{\Omega_e} \left[\mathbf{B}^T\mathbf{C}\mathbf{B} + \frac{l_1^2}{10}\mathbf{B}_{,x}^T\mathbf{C}\mathbf{B}_{,x} + \frac{l_2^2}{10}\mathbf{B}_{,y}^T\mathbf{C}\mathbf{B}_{,y} + \frac{l_3^2}{10}\mathbf{B}_{,z}^T\mathbf{C}\mathbf{B}_{,z}\right] d\Omega. \tag{76}$$

For a rotated ellipsoidal neighbourhood, the general tensor form in Eq. (48) is used.

The consistent mass matrix is

$$\mathbf{M}^e = \int_{\Omega_e} \rho\, \mathbf{N}_v^T\mathbf{N}_v\, d\Omega. \tag{77}$$

The mesh density adopted for the three-dimensional computations, together with its sensitivity, is quantified in Table B.3 (Appendix B).

5.5. Implementation Remarks

Hermite-type elements contain displacement derivatives as nodal degrees of freedom. The corresponding stiffness and mass matrices may therefore contain entries with different physical dimensions and scales. To improve conditioning, a symmetric diagonal scaling is applied to the free-degree system. For static analysis,

$$\tilde{\mathbf{K}} = \mathbf{S}\mathbf{K}_{ff}\mathbf{S}, \qquad \tilde{\mathbf{F}} = \mathbf{S}\mathbf{F}_f, \qquad \mathbf{d}_f = \mathbf{S}\tilde{\mathbf{d}}, \tag{78}$$

where $\mathbf{K}_{ff}$ and $\mathbf{F}_f$ are the stiffness matrix and force vector after imposing essential boundary conditions, $\tilde{\mathbf{K}}$ and $\tilde{\mathbf{F}}$ are the scaled stiffness and force, $\mathbf{d}_f$ is the physical displacement vector, $\tilde{\mathbf{d}}$ is the scaled displacement, and $\mathbf{S}$ is the diagonal scaling matrix. In the present implementation, $\mathbf{S}$ is chosen as a diagonal matrix based on the inverse square root of the positive diagonal entries of $\mathbf{K}_{ff}$.

For free-vibration analysis, the scaled generalized eigenvalue problem is

$$\tilde{\mathbf{K}}\tilde{\boldsymbol{\phi}} = \omega^2\tilde{\mathbf{M}}\tilde{\boldsymbol{\phi}}, \qquad \tilde{\mathbf{M}} = \mathbf{S}\mathbf{M}_{ff}\mathbf{S}, \tag{79}$$

where $\tilde{\mathbf{K}}$ and $\tilde{\mathbf{M}}$ are the scaled stiffness and mass matrices, $\mathbf{M}_{ff}$ is the mass matrix after imposing essential boundary conditions, $\tilde{\boldsymbol{\phi}}$ is the scaled eigenvector, and ω is the natural circular frequency, and the physical eigenvector is recovered from

$$\boldsymbol{\phi} = \mathbf{S}\tilde{\boldsymbol{\phi}}. \tag{80}$$

The scaling changes only the numerical representation of the algebraic system; it does not modify the finite element

approximation.

The essential boundary conditions in the direct strain-gradient implementation include classical displacement constraints and, when prescribed, Hermite derivative constraints. On boundaries where the corresponding derivative degrees of freedom are left free, the natural higher-order boundary conditions are enforced through the weak form. The present formulation is given first in general 3D form. The weak form is $\int \varepsilon(v)C\varepsilon(u) + 1/10 \int L_{mn}\varepsilon_{,m}(v)C\varepsilon_{,n}(u)$ with stiffness $K = K_{cl} + K_g$. The gradient part is $K_g = 1/10 \int \sum L_{\alpha\beta}(\partial B/\partial x_\alpha)^T C(\partial B/\partial x_\beta)$. One-dimensional cubic Hermite, two-dimensional BFS bicubic and three-dimensional tricubic elements are reduced kinematic specializations of the same 3D continuum, corresponding to uniaxial $u(x)$, plane-strain $u(x, y)$ and full $u(x, y, z)$. The 3D coefficient $1/10$ is retained for cross-dimensional consistency, as detailed in Appendix A. For truly lower-dimensional materials $1/6$ for 1D and $1/8$ for 2D would be appropriate. Extension to irregular geometries requires isoparametric C^1 Hermite elements. The parent domain $(\xi, \eta, \zeta) \in [-1, 1]^3$ is mapped to the physical domain via Hermite blending that preserves C^1 continuity. Derivatives are transformed via the Jacobian $J = \partial \mathbf{x}/\partial \boldsymbol{\xi}$ with the Hessian for second derivatives, and C^1 continuity of the normal derivative across curved interfaces is enforced via constraints or BFS blending. For general quadrilateral and hexahedral meshes C^1 continuity is non-trivial and requires Argyris or Bell elements, or alternatively isogeometric analysis with NURBS basis (C^{p-1}) or virtual element formulations [55]. The present rectangular formulation serves as a verification basis, with curved boundaries left as future work. For 3D mesh sensitivity, tricubic Hermite elements contain derivative DOFs. The clamped-gradient conditions $u_i = 0$ and $u_{i,j} = 0$ at $x = 0$ are enforced essentially, which allows an exponential boundary layer $\theta(x) \approx F/EA(1 - e^{-x/a})$ with $a = \sqrt{l^2/10} = 0.0158$ m for $l = 0.05$ m. For a $2 \times 1 \times 1$ mesh $h_x = 0.5$ m $\gg a$, the boundary layer is under-resolved. This causes 5.9% displacement variation and 9.3% frequency variation compared with a $6 \times 3 \times 3$ mesh with $h_x = 0.167$ m. Full results for all three meshes are reported in Table B.3 (Appendix B). The computed results (script `mesh_sensitivity_3D.m`) show displacement increasing from $u = 4.5665$ μm at $2 \times 1 \times 1$ to $u = 4.8371$ μm at $6 \times 3 \times 3$, while frequency decreases from $f_1 = 193.00$ Hz to $f_1 = 174.96$ Hz, with the intermediate $4 \times 2 \times 2$ mesh giving $u = 4.7876$ μm and $f_1 = 178.72$ Hz. Convergence is monotonic from below for displacement and from above for frequency, consistent with a conforming over-stiff formulation. Expressed as a resolution ratio, the three meshes give $h_x/a = 31.6$, 15.8 and 10.5 for $2\times1\times1$, $4\times2\times2$ and $6\times3\times3$ respectively, so none of them reaches the $h_x \ll a$ regime. The monotonic trend and the 5.9% and 9.3% spreads therefore bound the discretisation error associated with the boundary layer, but they do not eliminate it. The $4\times2\times2$ mesh used throughout Sec. 6 differs from the finest mesh tested by about 1.0% in displacement and 2.1% in frequency, and because the trend has not saturated these figures are lower bounds on the error remaining at that density. The element choice partially offsets the coarse resolution: since the tricubic Hermite element carries slope and curvature degrees of freedom, it can represent the exponential decay $\theta(x) \propto 1 - e^{-x/a}$ within a single element, whereas a C^0 element of the same size can reproduce that profile only piecewise linearly. That is a relative advantage at a given h_x/a, not a substitute for resolving the layer, and the boundary layer is not claimed to be fully resolved at any of the three densities. A fourth-order rate is expected only when $h_x \ll a$, which requires more than 20 elements in the x-direction for $l/L_x = 0.05$. Gradient energy is integrated with $3 \times 3 \times 3$ Gauss points exactly. Diagonal scaling keeps $\kappa < 10^6$ and equilibrium error $< 10^{-11}$. With respect to micro-inertia, the present model uses classical kinetic energy $T = 1/2 \int \rho \dot{u}_i \dot{u}_i d\Omega$ and omits $T_g = 1/2 \int \rho l_k^2 \dot{u}_{i,j} \dot{u}_{i,j} d\Omega$ [14]. Including T_g gives an enriched mass M_g and dispersion $\omega^2 = c^2k^2(1 + l_s^2k^2/10)/(1 + l_k^2k^2)$. For weakly nonlocal response $l_k k \ll 1$ and low-frequency first six modes with $k_n L \sim (2n - 1)\pi/2$, the condition $lk_n < 0.3$ holds for $l/L \leq 0.05$ and the effect is $< 5\%$, so it is omitted for clarity.

6. Numerical Results and Discussion

6.1. Material and Simulation Parameters

Unless otherwise specified, all numerical results presented in this study are obtained for a structural steel characterised by a Young's modulus of $E = 200$ GPa and a mass density of $\rho = 7800$ kg m^{-3}.

For adopted parameters for the one dimensional bar example are $L = 1$ m, $A = 1 \times 10^{-4}$ m^2, and $F = 10^4$ N. Whereas for a two dimensional plane-strain domain example a domain of size $L_x \times L_y = 1$ m $\times 0.2$ m is considered with the Poisson

ratio is $\nu = 0.30$ and uniform right edge traction $t_x = 10^6$ Pa. In order to examine size effects, the ratio l/L is chosen in the scale of 0 to 0.20. l_i are material constants (fabric-tensor-like second moments), while anisotropy ratio $AR = l_x/l_y$ and orientation θ are processing-induced design manifestations (fibre orientation, print direction, drawing direction), not optimization variables tuned with external fields. See Sec.7. A boundary layer reference case with $l/L = 0.05$ has been considered unless otherwise noted. These settings help to expose the exponential boundary layer profile and maintain a global displacement response.

The one-dimensional problem is treated first since the clamped slope strain gradient bar admits a closed-form analytical solution enabling rigorous verification of the proposed C^1 Hermite finite element implementation. Following the verification for one dimension, the formulation is extended to two and three dimensions via Bogner–Fox–Schmit (BFS) bicubic Hermite element, and to three-dimensional problems using the tensor-product tricubic Hermite element, respectively. These extensions preserves the same inter-element continuity conditions and the gradient stiffness constructions.

6.2. One-Dimensional Strain-Gradient Bar

6.2.1. Boundary-Layer Characteristics

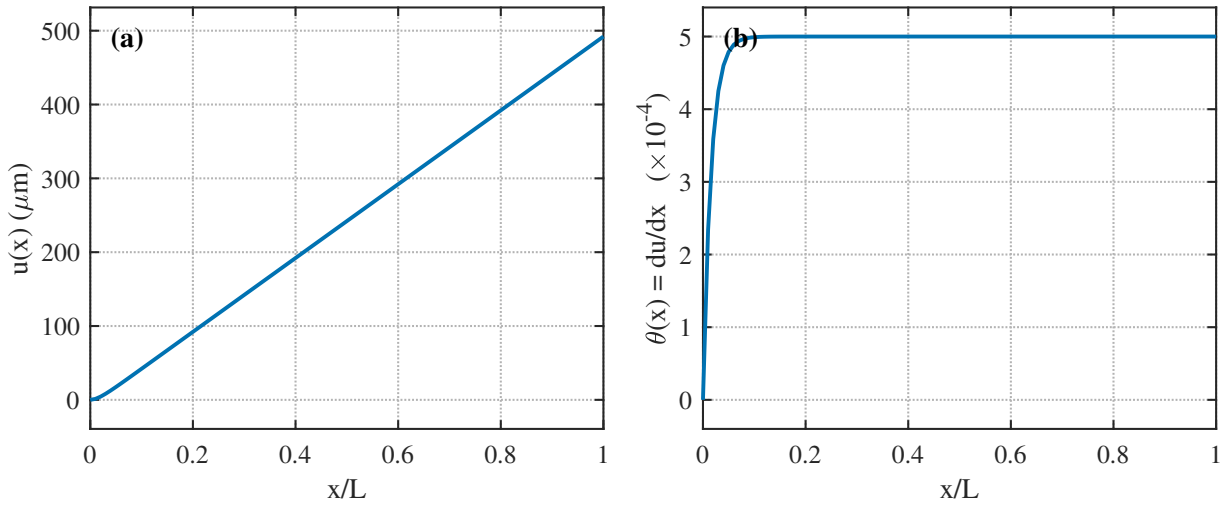


Figure 2: Boundary-layer characteristics of the one-dimensional strain-gradient bar under clamped-slope conditions ($l/L = 0.05$, $E = 2 \times 10^{11}$ Pa, $A = 10^{-4}$ m^2, $L = 1$ m, $F = 10^4$ N, $N_{\text{el}} = 100$): (a) axial displacement distribution $u(x)$; (b) slope distribution $\theta(x) = \mathrm{d}u/\mathrm{d}x$. Displacement remains nearly linear, while slope shows exponential layer near clamped end.

The displacements and their slopes predicted using the proposed C^1 Hermite finite element method for the strain gradient bar under tip load with essential boundary condition $\theta(0) = 0$ applied at the fixed end are shown in Figure 2. As can be seen from the two plots, the solution uses two different length scales: a globally nearly linear displacement that is determined by the classical stiffness, and an exponentially decaying local correction to the strain field in a thin layer near the clamped end.

It can be seen from panel (a) that the displacement is a monotonic function, starting from $u(0) = 0$ up to the end subjected to loading. At the scale of the entire domain the curve is visually indistinguishable from the classical one, $u_{\text{cl}}(x) = Fx/(EA)$, as the axial force is everywhere constant and thus would produce uniform strain in the case of no additional constraints. Analytical value of the displacement at the loaded end in the case of a clamped-slope problem subjected to a tip force is $u_{\text{SG}}(L) = (F/EA)(L - a\tanh(L/a))$, resulting in an approximate value of $492\,\mu$m — some 1.6 % less than the classical one, $FL/(EA) = 500\,\mu$m.

The effect cannot be observed on the scale of panel (a), but it is important in that it constitutes an observable effect of the microstructural length parameter: it is non-zero for any $l > 0$ and is missing in any classical analysis, regardless of the fineness of the mesh used. The observed stiffening is a direct consequence of the higher-order slope constraint $\theta(0) = 0$, which forces the strain distribution to depart from uniformity near the fixed end, thereby activating the strain-gradient

energy contribution. If natural boundary conditions are imposed (i.e., no constraint on the slope at either end), then the tip-loaded strain gradient bar has the classical uniform strain solution as its exact solution.

Slope distribution depicted in panel (b) illustrates this process directly. According to classical theory of elasticity, axial strain $\varepsilon = \mathrm{d}u/\mathrm{d}x$ under a constant internal resultant of force is uniform and equals $F/(EA)$. The second-order governing equation does not have a separate essential boundary condition for slope and cannot generate any boundary layer. The fourth-order governing equation with respect to the displacement allows imposing two essential boundary conditions independently of the displacement boundary conditions. The slope boundary condition $\theta(0) = 0$ corresponds precisely to this case: it is possible to impose this condition in the context of strain gradient theory but has no counterpart in classical theory.

The slope grows from zero value at $x = 0$, goes through the sharp monotonic increase and approaches the classical plateau $F/(EA) = 5 \times 10^{-4}$ when $x/L \lesssim 0.10$. Then, $\theta(x)$ becomes constant to the precision of the graph plotting for the rest of the area.

The length of the boundary layer is determined by the gradient length $a = \sqrt{l^2/10}$ resulting from the ellipsoidal non-local averaging procedure.

$$\theta(x) \approx \frac{F}{EA}\left(1 - e^{-x/a}\right), \tag{81}$$

Given the current parameters, $a/L \approx 1.6 \times 10^{-2}$, which means that the layer covers approximately the first two percent of the total length of the bar. Given the assumption that $L/a \gg 1$, which applies for the current case ($L/a \approx 63$), the exact analytical solution reduces to the one which shows that the boundary layer is governed by an exponential kernel with length scale equal to a.

The classical result is obtained in the limiting case of $l \to 0$, as $a \to 0$ implies $e^{-x/a} \to 0$ for all $x > 0$. The finite element results agree with the analytical slope profile to 10^{-3} % accuracy in the case of the present 100-element mesh; the relative tip-displacement error reduces from about 1.2 % in the case of a coarse 5-element mesh to less than 10^{-3} % in the present 100-element mesh, indicating the fourth order of algebraic convergence of the C^1 cubic Hermite interpolation which is expected and proving that the layer structure is well resolved with a uniformly-spaced mesh when $h \ll a$.

Together, both panels show that the strain-gradient influences act within the boundary zones due to higher-order boundary conditions, whereas the interior behavior follows classic elasticity principles. This separation of scales is an inherent property of Aifantis-type gradient elastic models. Such behavior of solutions leads us to the parametric study of the microstructural parameter.

6.2.2. Influence of the Microstructural Length Parameter on Static Response

Figure 3 presents the results of a parametric study in which the microstructural length parameter l is varied over the range $l/L \in [0, 0.20]$ while all other material and geometric parameters are held constant. Panel (a) demonstrates that the tip displacement $u(L)$ decreases monotonically as l/L increases, departing progressively from the classical prediction $u_{\mathrm{cl}}(L) = FL/(EA) = 500\,\mu$m and reaching approximately $468\,\mu$m at $l/L = 0.20$. This behavior shows that this stiffening is not a change in the classical elastic modulus, but the activation of a higher-order resistance to non-uniform strain distributions. The clamped-slope condition $\theta(0) = 0$ forces the axial strain to depart from the uniform state near the fixed boundary, and the strain-gradient energy term

$$\tfrac{1}{2} EA\, a^2 \int_0^L (u'')^2 \,\mathrm{d}x$$

penalises the resulting strain curvature with a contribution that scales with $a^2 = l^2/10$. This additional stiffness grows with the microstructural length parameter l, which characterises the spatial extent of the nonlocal microstructural averaging domain and is a property of the material rather than of the bar geometry. Consequently, the effective structural stiffness

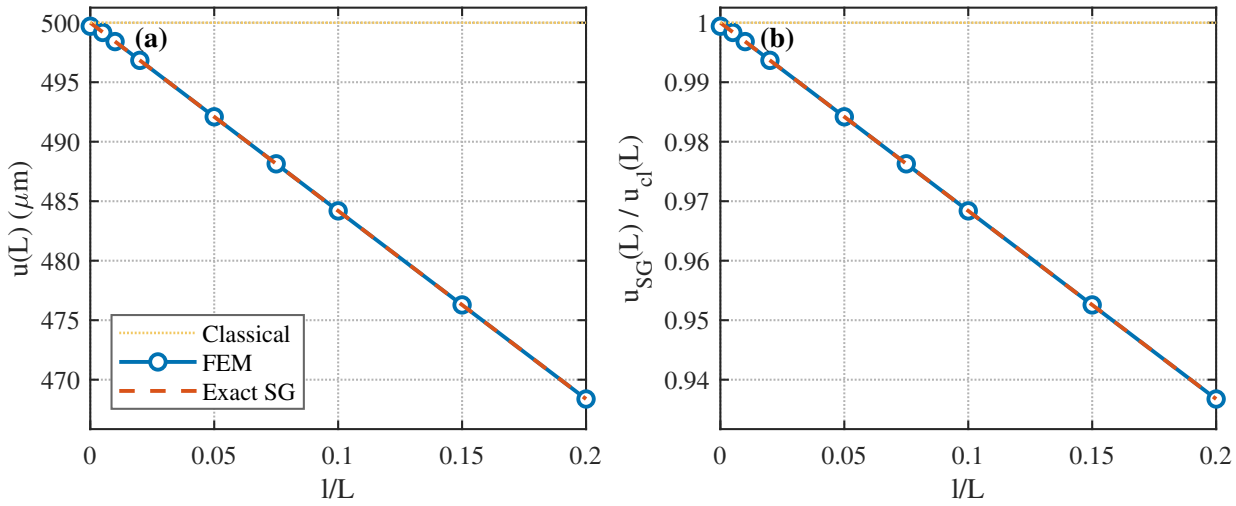


Figure 3: Influence of the microstructural length parameter on the static response of the one-dimensional strain-gradient bar ($E = 2 \times 10^{11}$ Pa, $A = 10^{-4}$ m^2, $L = 1$ m, $F = 10^4$ N): (a) tip displacement $u(L)$; (b) normalised displacement ratio $u_{\mathrm{SG}}(L)/u_{\mathrm{cl}}(L)$ versus l/L. Both quantities decrease monotonically with l/L.

increases with l, and the corresponding tip displacement falls below the classical prediction by an amount that grows with l/L.

The normalised displacement ratio $u_{\mathrm{SG}}(L)/u_{\mathrm{cl}}(L)$ shown in panel (b) provides a dimensionless measure of the stiffening that is independent of the applied load F and the cross-sectional stiffness EA, depending instead solely on the dimensionless ratio l/L. The ratio equals unity at $l/L = 0$, confirming that the strain-gradient model converges to classical elasticity when the microstructural length vanishes, and it decreases with a near-linear dependence on l/L throughout the range considered. This near-linearity follows from the analytical solution $u_{\mathrm{SG}}(L) = (F/EA)(L - a\tanh(L/a))$: for $L/a \gg 1$ (satisfied for all values shown), $\tanh(L/a) \to 1$ and the ratio reduces to $1 - l/(L\sqrt{10})$, a linear function of l/L with slope $-1/\sqrt{10}$. At $l/L = 0.10$, the normalised ratio is approximately 0.968, indicating a tip displacement roughly 3.2 % below the classical value — a difference that is systematic, repeatable, and absent from any classical finite element analysis regardless of mesh refinement. The agreement between the C^1 Hermite finite element solution and the exact analytical result is excellent in both panels, consistent with the fourth-order convergence established in Section 6.2.1.

These results confirm that the proposed one-dimensional discretisation is both sufficiently accurate and mesh-converged. Having established the size-dependent static response, we now examine the free-vibration characteristics of the strain-gradient bar.

6.2.3. Influence of the Microstructural Length Parameter on Free-Vibration Characteristics

Figure 4 presents the free-vibration response of the strain-gradient bar as a function of the normalised microstructural length l/L over the range [0, 0.20]. Panel (a) shows that every natural frequency increases monotonically with l/L, so the strain-gradient formulation predicts a stiffer dynamic response than classical elasticity for all modes and all non-zero values of the microstructural length. The increase is markedly mode-dependent across the spectrum. Notably Mode 1 increases only modestly from nearly 1.3 kHz to 1.4 kHz — whereas Mode 6 experiences a substantial elevation, growing from roughly 14 kHz to more than 21 kHz over the same parameter range. The growth patterns become increasingly nonlinear as l/L increases, with the modal patterns bending more sharply at larger values of the microstructural length. This concave-upward behaviour of the curves is consistent with the quadratic scaling of the strain-gradient contribution with l. Since the gradient energy term scales with $a^2 = l^2/10$, doubling l/L increases the gradient-energy contribution by a factor of four, thereby producing the increasingly nonlinear frequency growth observed in Figure 4(a). The monotonic increase of all natural frequencies is an inherent property of the present strain-gradient formulation. This follows from the fact that the gradient energy functional contributes a positive-definite increment to the stiffness matrix, while leaving the mass matrix unchanged. As the closed form eigen frequencies are not available for the clamped-slope strain-gradient bar,

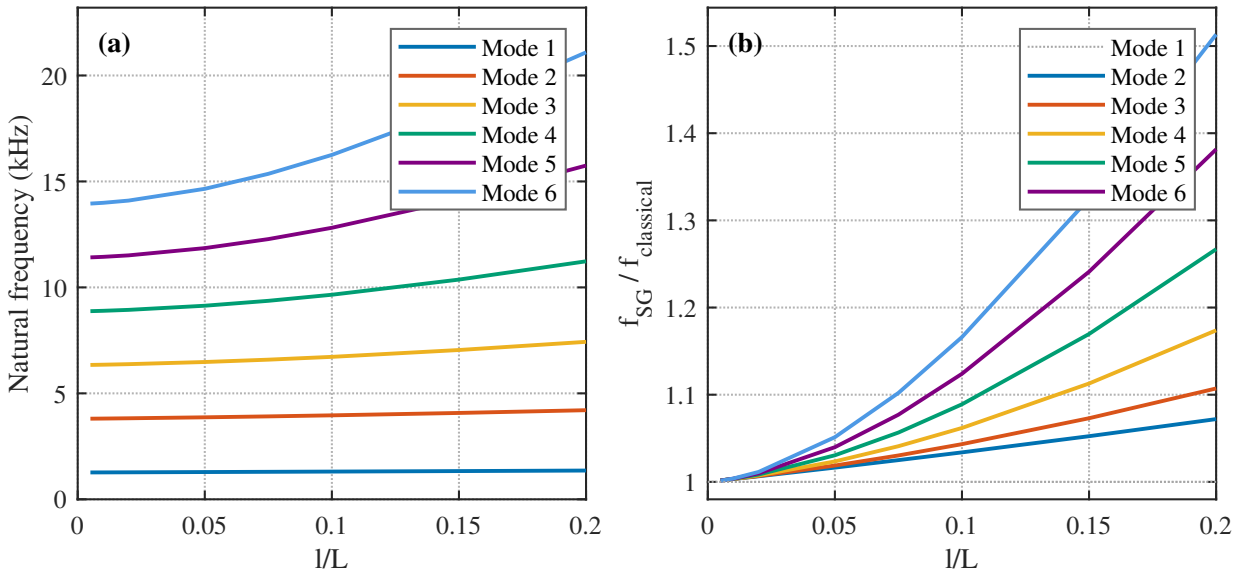


Figure 4: Free-vibration characteristics of the one-dimensional strain-gradient bar ($E = 2{\times}10^{11}$ Pa, $\rho = 7800\,\mathrm{kg\,m^{-3}}$, $A = 10^{-4}\,\mathrm{m^2}$, $L = 1$ m, $N_{\mathrm{el}} = 120$): (a) first six natural frequencies versus l/L; (b) normalised frequency ratio $f_{\mathrm{SG}}/f_{\mathrm{classical}}$ versus l/L. Every natural frequency increases monotonically with l/L, with higher modes exhibiting progressively greater strain-gradient stiffening.

the dynamic verification is established on the internal spectrum consistency and on the correct limiting nature as $l \to 0$.

The mode-dependent sensitivity apparent in panel (a) is further quantified in panel (b) by the means of the normalised frequency ratio $f_{\mathrm{SG}}/f_{\mathrm{classical}}$. The curve hierarchy is identical in both panels. Although the ordering of the curves is identical in both panels, their physical interpretation is different. Higher modes carry larger absolute frequencies in panel (a), whereas panel (b) shows that they also experience proportionally larger frequency gains. Mode 6 reaches a ratio of approximately 1.52 at $l/L = 0.20$, while Mode 1 retains the smallest ratio of roughly 1.08 at the same length. This mode-selective amplification arises from the spectral structure of the gradient energy: for mode n of the clamped bar the modal wavenumber scales as $k_n \propto (2n-1)$, and the gradient energy contribution to the n-th Rayleigh quotient involves $(a\,k_n)^2$, which grows as $(2n-1)^2$. Because this quantity increases rapidly with n, the gradient energy represents a progressively larger fraction of the total elastic energy for higher modes: it is nearly negligible for Mode 1 but dominant for Mode 6 when l/L is appreciable. High-frequency, short-wavelength modes are therefore far more sensitive to the microstructural length than their low-frequency counterparts, a behaviour that has no counterpart in scale-independent classical continuum mechanics.

At the smallest values of l/L shown, all six ratio curves originate near unity, confirming that the strain-gradient model approaches the classical solution as $l \to 0$. The fact that the ratios are slightly above unity even at the smallest l/L reflects the combined contribution of the gradient energy and the additional kinematic boundary condition $\theta(0) = 0$, which has no counterpart in the classical model and independently raises all natural frequencies by constraining an additional degree of freedom. As l/L grows, the gradient energy term dominates and the mode-dependent amplification accelerates, producing the fan-shaped divergence of the ratio curves evident in panel (b).

The one-dimensional study has established that the proposed C^1 Hermite formulation accurately reproduces the static strain-gradient response through comparison with the analytical solution, while the dynamic formulation exhibits the correct limiting behaviour and internally consistent spectral characteristics. The natural progression is to generalize the formulation to two dimensions, allowing the full ellipsoidal nonlocal operator with independent characteristic lengths l_x and l_y to be activated and the anisotropic stiffening predicted by the theory to be evaluated in quantitative terms.

6.3. Two-Dimensional Plane-Strain BFS Formulation

The generalization to two dimensions employs the Bogner–Fox–Schmit (BFS) bicubic Hermite rectangular element, which provides the C^1 inter-element continuity required by the fourth-order strain-gradient operator. The plane-strain

constitutive matrix and the gradient-stiffness assembly follow directly from Section 5.3. Because no closed-form analytical solution exists for the two-dimensional clamped-gradient problem, the BFS implementation is verified through mesh convergence (Section 6.3.2) and through cross-comparison with the one-dimensional results where the physics permits.

6.3.1. *Influence of the Microstructural Length Parameter on Static Response*

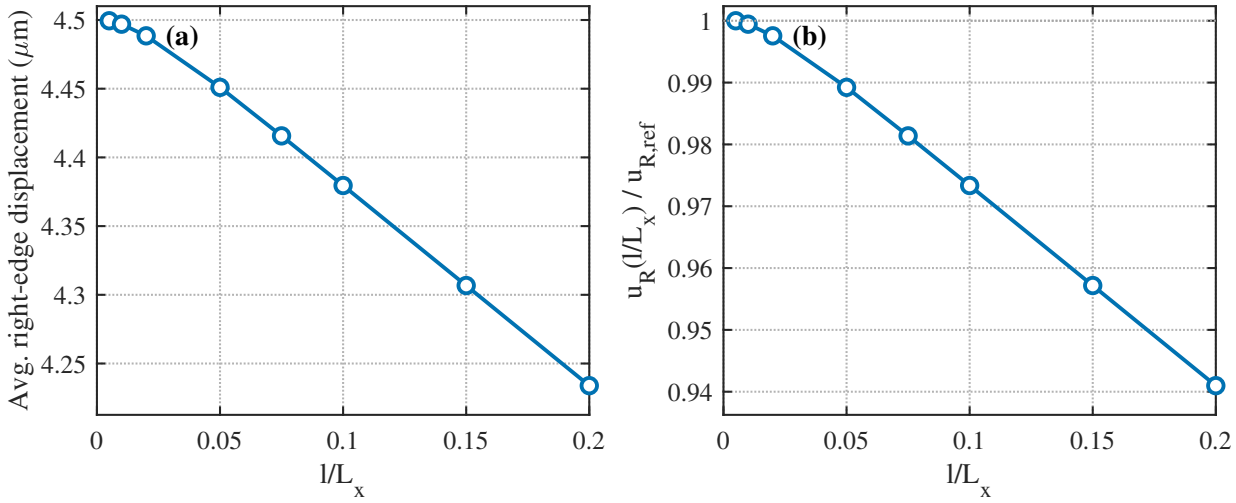


Figure 5: Influence of the microstructural length parameter on the static response of the two-dimensional strain-gradient plane-strain domain ($E = 2 \times 10^{11}$ Pa, $\nu = 0.30$, $L_x = 1$ m, $L_y = 0.2$ m, $t_x = 10^6$ Pa, 20×4 BFS mesh): (a) average right-edge displacement $\bar{u}_{\mathrm{R}}$; (b) normalised displacement ratio $\bar{u}_{\mathrm{R}}/\bar{u}_{\mathrm{R,ref}}$ versus l/L_x. Both quantities decrease monotonically with l/L_x.

Figure 5 extends the microstructural length-scale study of Section 6.2.2 to the two-dimensional plane-strain setting, using the BFS formulation on a 20×4 rectangular mesh. Panel (a) shows that the average right-edge displacement $\bar{u}_{\mathrm{R}}$ decreases monotonically as l/L_x increases from the smallest value considered to $l/L_x = 0.20$. The displacement decreases from approximately 4.50 μm to approximately 4.23 μm, corresponding to a reduction of nearly 6 % relative to the reference configuration. The profile exhibits a smooth and nearly linear decline over the range considered, indicating a systematic increase in strain-gradient stiffening with increasing microstructural length. The physical origin of this behaviour is identical to that observed in the one-dimensional study: the clamped-gradient boundary conditions generate non-uniform strain fields near the constrained boundary, thereby activating the gradient energy contributions. As the microstructural length increases, these higher-order energy terms contribute more significantly to the total strain energy, resulting in an increase in the effective structural stiffness and a corresponding reduction in deformation under the same applied loading.

Panel (b) isolates the size effect through the normalised displacement ratio $\bar{u}_{\mathrm{R}}(l/L_x)/\bar{u}_{\mathrm{R,ref}}$, where $\bar{u}_{\mathrm{R,ref}}$ denotes the displacement corresponding to the smallest microstructural length considered. The ratio approaches unity as $l/L_x \to 0$ and decreases monotonically to approximately 0.94 at $l/L_x = 0.20$. A comparison with the one-dimensional results of Section 6.2.2 reveals a remarkably similar level of stiffening, despite the additional kinematic constraints and two-directional gradient effects present in the plane-strain formulation. This close agreement confirms that the microstructural length parameter governs the dominant stiffening mechanism in a consistent manner across both formulations, and that the l/L scaling established in one dimension remains a reliable predictor of the order of magnitude of size effects in two dimensions. The smoothness of the computed response, the absence of spurious oscillations, and the physically consistent limiting behaviour as $l/L_x \to 0$ collectively provide preliminary evidence that the BFS discretisation correctly captures the governing strain-gradient mechanics.

Before proceeding to the anisotropic investigation, the mesh independence of the BFS formulation must be established.

6.3.2. *Mesh Convergence of the BFS Formulation*

Figure 6 examines the spatial convergence of the two-dimensional strain-gradient formulation through systematic mesh refinement of the BFS discretisation. The average right-edge displacement $\bar{u}_{\mathrm{R}}$ is monitored while maintaining a constant

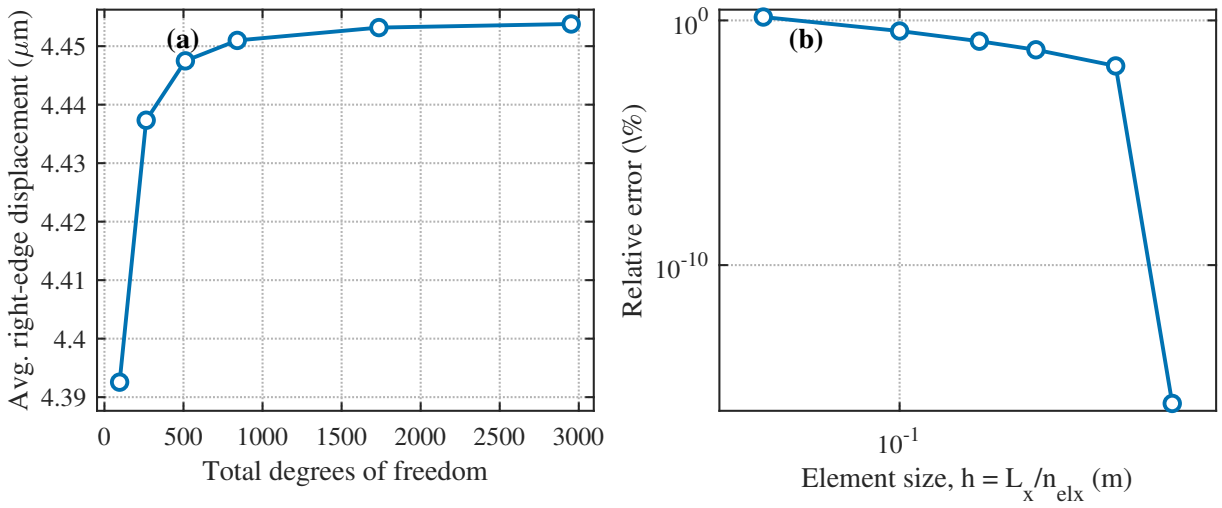


Figure 6: Mesh convergence of the two-dimensional plane-strain BFS formulation ($E = 2 \times 10^{11}$ Pa, $\nu = 0.30$, $L_x = 1$ m, $L_y = 0.2$ m, $l/L_x = 0.05$, $t_x = 10^6$ Pa): (a) average right-edge displacement $\bar{u}_R$ versus total degrees of freedom; (b) relative error in $\bar{u}_R$ versus element size h. Displacement converges monotonically; error decreases near fourth-order rate.

element aspect ratio $n_{elx}/n_{ely} = L_x/L_y = 5$. Six meshes ranging from 5×1 to 40×8 elements are considered, corresponding to 96 to 2952 total degrees of freedom. The study is performed at $l/L_x = 0.05$, the reference microstructural length used in the preceding parametric investigation.

Panel (a) illustrates the variation of the average right-edge displacement with increasing mesh density. The displacement increases monotonically from approximately 4.39 μm on the coarsest mesh to approximately 4.45 μm on the finest mesh, gradually approaching a stable limiting value. The monotone increase, with $\bar{u}_R$ approaching its converged value from below, is the over-stiff behaviour characteristic of a conforming displacement-based finite element formulation: by the Rayleigh–Ritz principle the discrete solution is at least as stiff as the exact one on any given mesh, and the gap narrows as the mesh is refined. The difference between the adopted 20×4 mesh and the finest reference mesh is less than 0.07 %, indicating that the computed displacement response has effectively reached mesh independence and that the size-dependent stiffening trends reported in Figure 5 are genuine physical effects rather than numerical artefacts of insufficient spatial resolution.

Panel (b) quantifies the convergence rate through the relative error, referenced to the finest mesh. The error falls from approximately 1.4 % at the coarsest discretisation to below 0.1 % by the 20×4 mesh and below 0.02 % at 30×6. On the log–log scale the slope steepens progressively, approaching the fourth-order algebraic rate expected of the C^1 cubic Hermite interpolation; the reduced apparent order on the coarsest meshes reflects incomplete resolution of the boundary layer when the element size h exceeds the gradient length a by roughly an order of magnitude. This behaviour is consistent with the one-dimensional convergence established in Section 6.2.1.

Overall, the results provide clear evidence that the two-dimensional implementation is spatially converged and numerically reliable. The 20×4 discretisation offers an effective balance between computational efficiency and numerical accuracy, and is therefore employed in all subsequent two-dimensional parametric studies.

6.3.3. *Influence of the Anisotropic Characteristic Lengths*

Having verified the spatial convergence of the BFS formulation in Section 6.3.2, the present subsection examines the direction-dependent stiffening induced by anisotropic characteristic lengths. Whereas the study of Section 6.3.1 employed a single isotropic length $l_x = l_y = l$, the governing equations admit distinct characteristic lengths in the axial and transverse directions through the gradient operator

$$L = l_x^2 \frac{\partial^2}{\partial x^2} + l_y^2 \frac{\partial^2}{\partial y^2},$$

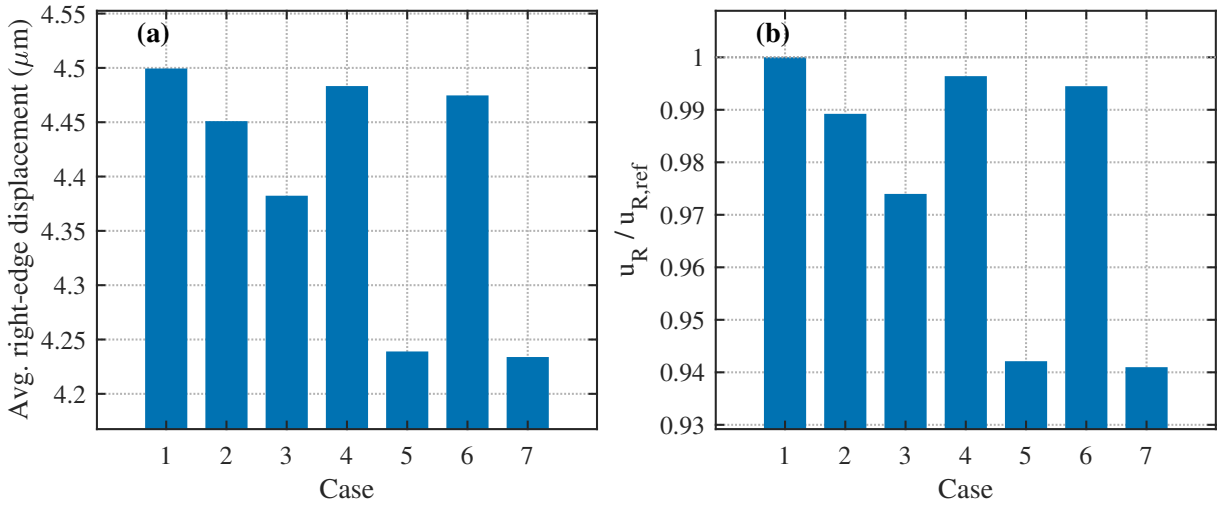


Figure 7: Influence of the anisotropic characteristic lengths on the static response of the two-dimensional strain-gradient plane-strain domain ($E = 2 \times 10^{11}$ Pa, $\nu = 0.30$, $L_x = 1$ m, $L_y = 0.2$ m, $t_x = 10^6$ Pa, 20×4 BFS mesh): (a) average right-edge displacement $\bar{u}_\mathrm{R}$ for seven (l_x, l_y) combinations; (b) normalised displacement ratio $\bar{u}_\mathrm{R}/\bar{u}_\mathrm{R,ref}$. Larger axial l_x gives greater stiffening.

and the associated strain-gradient energy contribution

$$\frac{1}{20}\Big[l_x^2(\partial_x \boldsymbol{\varepsilon})^\mathsf{T}\mathbf{C}\,(\partial_x \boldsymbol{\varepsilon}) + l_y^2(\partial_y \boldsymbol{\varepsilon})^\mathsf{T}\mathbf{C}\,(\partial_y \boldsymbol{\varepsilon})\Big].$$

Seven representative (l_x, l_y) combinations are considered on the same 20×4 mesh, ranging from the near-classical configuration $(0.005, 0.005)$ to strongly anisotropic cases in which one characteristic length is an order of magnitude larger than the other.

Panel (a) of Figure 7 shows that the displacement response depends strongly on which characteristic length is increased. The near-classical case $(0.005, 0.005)$ yields $\bar{u}_\mathrm{R} \approx 4.50\,\mu$m, while the isotropic configuration $(0.05, 0.05)$ produces $\bar{u}_\mathrm{R} \approx 4.45\,\mu$m, consistent with the stiffening established in Figure 5. The influence of anisotropy becomes evident when comparing transposed pairs of characteristic lengths. The configuration $(0.10, 0.02)$ yields $\bar{u}_\mathrm{R} \approx 4.38\,\mu$m, whereas the transposed case $(0.02, 0.10)$ yields $\bar{u}_\mathrm{R} \approx 4.48\,\mu$m, corresponding to a difference of approximately 2.3 % despite involving the same set of length magnitudes. The disparity becomes more pronounced at larger values, where the pair $(0.20, 0.02)$ and $(0.02, 0.20)$ differs by approximately 5.3 %. In both comparisons, the configuration with the larger axial characteristic length l_x exhibits the lower displacement and therefore the greater effective stiffness. A large transverse length alone, $(l_x, l_y) = (0.02, 0.20)$, reduces the displacement by only about 0.5 % relative to the reference, whereas the axial-dominant case $(0.20, 0.02)$ produces a reduction of approximately 5.8 %.

The origin of this directional sensitivity lies in the interaction between the strain-gradient energy and the dominant deformation mode. Under the applied right-edge traction, the primary strain component is the axial strain $\varepsilon_{xx} = \partial u/\partial x$. The clamped-gradient condition at the left boundary generates the same type of boundary layer identified in the one-dimensional study (Section 6.2.1), in which significant axial strain gradients develop near the constrained edge. The energy contribution weighted by l_x^2 directly penalises this dominant strain variation and increases rapidly as l_x grows. By contrast, the transverse strain-gradient contribution associated with l_y^2 remains comparatively small because the plane-strain domain is slender ($L_y/L_x = 0.2$) and the deformation is predominantly axial. This explains why the fully enlarged configuration $(0.20, 0.20)$ yields $\bar{u}_\mathrm{R} \approx 4.23\,\mu$m and a normalised ratio of approximately 0.94 — essentially identical to the axial-dominant case $(0.20, 0.02)$ at 0.942: once l_x becomes sufficiently large, further increasing l_y has only a minor influence on the global response. This result also provides a consistency check, since the isotropic $(0.20, 0.20)$ case reproduces the ratio reported in Figure 5 at $l/L_x = 0.20$.

Panel (b) presents the same trends through the normalised displacement ratio $\bar{u}_\mathrm{R}/\bar{u}_\mathrm{R,ref}$, thereby removing the dependence on the absolute displacement scale. The ratios for the axial-dominant cases decrease to approximately 0.974 and

0.942, whereas those for the transverse-dominant cases remain close to unity at approximately 0.997 and 0.995. The separation between these groups increases systematically as the characteristic lengths grow, demonstrating that the effective strain-gradient stiffening is governed primarily by the characteristic length aligned with the loading direction.

Considering the engineering point of view, These findings demonstrate practical value beyond isotropic models. (i) Axial versus transverse dominance is clear. The pair $(0.10, 0.02)$ versus $(0.02, 0.10)$ differs by 2.3%, while $(0.20, 0.02)$ versus $(0.02, 0.20)$ differs by 5.3%. Axial-dominant $(0.20, 0.02)$ produces 5.8% reduction, whereas transverse-dominant $(0.02, 0.20)$ gives only 0.5%. An isotropic assumption would therefore under-predict axial stiffening and over-predict transverse response, which is critical for fibre composites where $l_x \gg l_y$. (ii) The design space in Fig.11 reveals three regions. These are stiffening-optimal at high AR and low θ with max DSI, frequency-tuning at high AR near $\theta \approx 45°$, and insensitive when AR$\lesssim 2$ or $\theta \approx 90°$. (iii) DSI is defined as $DSI = (u_{ref} - u)/u_{ref}$ with u_{ref} from volume-equivalent isotropic $l_{iso} = (l_x l_y l_z)^{1/3}$. It quantifies displacement reduction versus the isotropic baseline and enables tailoring by aligning the major axis with the load for maximum stiffness or at 45° for minimum frequency. Values from the corrected solver use $l_{iso} = (l_x l_y l_z)^{1/3} = 0.0431$ m. With $u_{ref} = 4.80$ μm, $u(0°) = 4.6137$ μm gives DSI= 3.9% and $u(45°) = 4.7031$ μm gives DSI= 2.0%. The index is predominantly positive from 0% to +29.6%, with only small negative values of -0.1% to -0.9% at high orientation angles $\theta \gtrsim 75°$ for $AR \geq 2$. This confirms no -8% cross-term softening at 45° and confirms monotonic tuning. these findings indicate that l_x and l_y should be considered as independent material properties in microstructured media having directional morphology. In materials like fibre composite, layered film, or drawn metal, the primary effect could have a high dependence on the orientation of the characteristic microstructure environment. If we assume isotropic behaviour in such cases, there will be an underestimate of the stiffening effect in the dominant direction and, at the same time, an overestimate of the transverse direction. The spatial distribution of the displacement field causing the above mentioned effects will be discussed in the subsequent contour analysis.

6.3.4. Spatial Distribution of the Displacement Field

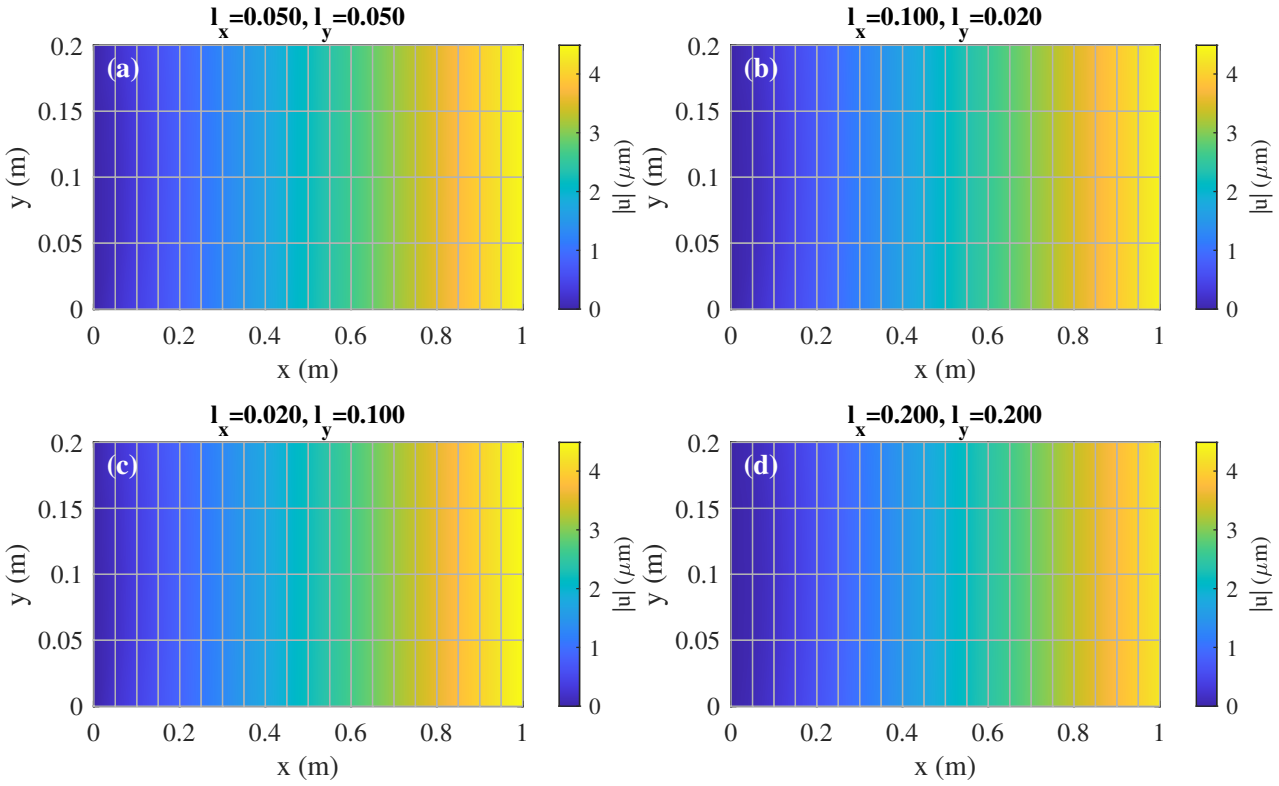


Figure 8: Spatial distribution of the displacement-magnitude field $|u|$ (μm) for four characteristic-length combinations ($E = 2 \times 10^{11}$ Pa, $\nu = 0.30$, $L_x = 1$ m, $L_y = 0.2$ m, $t_x = 10^6$ Pa, 20×4 BFS mesh, shared colour scale): (a) isotropic $(0.05, 0.05)$; (b) x-dominant $(0.20, 0.02)$; (c) y-dominant $(0.02, 0.20)$; (d) both large $(0.20, 0.20)$. Panels (b) and (d) nearly coincide, confirming axial dominance.

Integrated displacement ratios shown in Figure 7 have indicated that the stiffening of the plane-strain body depends mainly on the axial characteristic length l_x. Figure 8 illustrates this conclusion geometrically in terms of the distribution of the displacement magnitude field $|u|$ for four representative values of (l_x, l_y) subjected to the same loading and mesh. In each subfigure, the field monotonically grows from zero at the clamped boundary at $x = 0$ up to the maximum value at the loaded boundary. Since the deformation occurs uniformly along the tensile loading direction x, $|u|$ shows no significant variation through the thickness of the plane-strain body and it is less than 0.6 % of the magnitude. Consequently, $|u|$ can be regarded almost as horizontal bands with different colors, although it is computed using two-dimensional gradient energy formulation. The small variation through the thickness of the plane-strain body is due to the slender shape ($L_y/L_x = 0.2$) of the plane-strain body and the axial loading.

The very important aspect which may be achieved from the figure is the coupled comparison of the four panels. Both the Panels namely (b) and (d); the x-dominant case $(0.20, 0.02)$ and the both-large case $(0.20, 0.20)$ are appeared to be visually nearly similar on the shared colour scale even though the transverse length differs by an order of magnitude. Quantitatively their displacement fields differ by nearly 0.3 % pointwise. On the contrast, for the panels namely (a) and (c) which are the isotropic case and the y-dominant case are likewise very similar as for both, the axial length remains small ($l_x \leq 0.05$) and the stiffening magnitude of l_y is minimal. The two groups are separated by a noticeable reduction in displacement, with the large-l_x group reaching a lower maximum ($|u|_{\max} \approx 4.24\,\mu$m) than the small-$l_x$ group ($|u|_{\max} \approx 4.45$–$4.48\,\mu$m). This observed accumulation offers a direct field-level indication of the integrated outcomes, depicting that the deformation field is governed by the l_x, not l_y and also as l_x becomes large, the effect of transverse characteristic length becomes weak.

The influence of the axial length on the near-edge field structure is consistent with the boundary-layer mechanism identified in the one-dimensional study (Section 6.2.1). In the isotropic panel (a), the displacement rises comparatively sharply over the first element adjacent to the clamped edge, consistent with the rapid strain recovery characteristic of the gradient boundary layer. In the x-dominant panel (b), this near-edge rise is visibly more gradual, indicating that the large l_x-weighted gradient energy smooths the axial strain distribution at the constrained boundary. The y-dominant panel (c) exhibits a response much closer to the isotropic case because the small value of l_x imposes only a weak penalty on the axial strain gradient. These near-edge differences remain confined to a narrow region; over most of the domain the field recovers an approximately linear profile in all four cases, consistent with the uniform-strain regime that prevails far from the constrained boundary.

It should be noted that the larger characteristic-length values considered in the present design-space study are primarily intended to illustrate parametric anisotropy trends. Accordingly, these results should be interpreted as a sensitivity analysis rather than as a strict asymptotic realization of the weakly nonlocal regime assumed in the theoretical development. Overall, the contour plots verify that the stiffness behaviour predicted by the anisotropic strain-gradient model in terms of its direction dependency is not only a consequence of integration but also a result of the geometry of the displacement field, with the characteristic length in the main deformation direction determining the distribution of the strain field adjacent to the clamped boundary, and, thus, the compliance of the plane-strain body. Therefore, the internal consistency between the 2D plane-strain static stiffening analysis (Section 6.3.1), mesh refinement analysis (Section 6.3.2), anisotropic length effects (Section 6.3.3) and the present field visualization is established at both local and global levels. This provides additional numerical support for the BFS formulation and establishes a consistent foundation for its extension to three dimensions.

6.4. Three-Dimensional Tricubic Hermite Formulation

The three-dimensional generalization uses the tensor-product tricubic Hermite hexahedral element, which allows for C^1 continuity between elements in all three coordinate directions that the fourth-order strain-gradient operator requires. The degrees of freedom at each node include 24 (the displacement, its three first-order partials, three second-order partials, and the mixed partial third derivative, of the three displacement components), and gradient stiffness assembly proceeds directly from the one- and two-dimensional cases. The relatively coarse $4 \times 2 \times 2$ grid used here accounts for the much greater computational expense of the three-dimensional approach while still being fine enough to capture the physics.

This applies to the global static and dynamic response and should not be read as a claim that the near-boundary strain-gradient layer is fully resolved: for $l = 0.05\,\mathrm{m}$ the layer thickness is $a = \sqrt{l^2/10} = 0.0158\,\mathrm{m}$, so the $4 \times 2 \times 2$ grid corresponds to $h_x/a \approx 15.8$. The mesh-sensitivity study summarised in Table B.3 and discussed in Sec. 5.5 quantifies the resulting discretisation error. As was true in two dimensions, no closed-form solution is available for the three-dimensional clamped gradient formulation, and the analysis is confirmed via comparison to the lower dimensional solutions as well as examination of the displacement fields.

6.4.1. Impact of the Microstructural Length Parameter on the Static Response

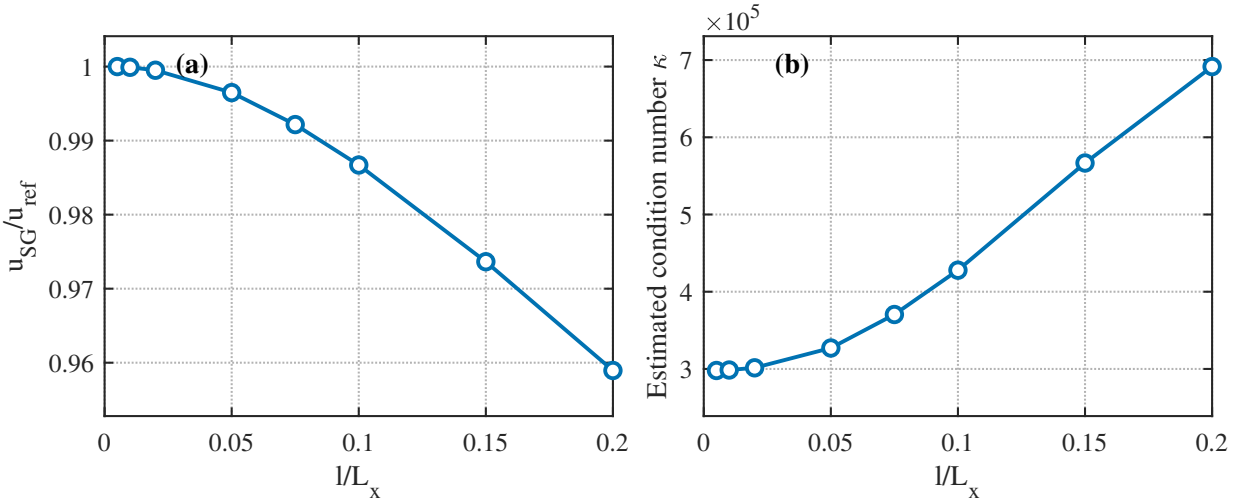


Figure 9: Influence of the microstructural length parameter on the three-dimensional strain-gradient cuboid ($E = 2 \times 10^{11}\,\mathrm{Pa}$, $\nu = 0.30$, $L_x = 1\,\mathrm{m}$, $L_y = L_z = 0.2\,\mathrm{m}$, $t_x = 10^6\,\mathrm{Pa}$, $4 \times 2 \times 2$ tricubic Hermite mesh): (a) normalised displacement ratio $u_{\mathrm{SG}}/u_{\mathrm{ref}}$ versus l/L_x; (b) estimated condition number κ versus l/L_x. Displacement ratio decreases monotonically; condition number remains moderate.

Figure 9 presents the extension of the microstructural length-scale investigation to the fully three-dimensional case, employing the tensor-product tricubic Hermite hexahedral formulation on a cuboidal domain $L_x \times L_y \times L_z = 1 \times 0.2 \times 0.2\,\mathrm{m}^3$ subjected to uniform traction of the right boundary. To examine the size-dependent response, the characteristic length parameter l is varied over the range $l/L_x \in [0,\ 0.20]$, with $l/L_x = 0.005$ serving as the near-classical reference case. This study serves two purposes: it confirms that the size-dependent strain-gradient stiffening established in the one- and two-dimensional analyses persists in three dimensions, and it documents the numerical conditioning of the tricubic Hermite formulation as the gradient parameter is increased.

Panel (a) shows that the normalised displacement ratio $u_{\mathrm{SG}}/u_{\mathrm{ref}}$ decreases monotonically as l/L_x increases, departing progressively from unity and reaching approximately 0.959 at $l/L_x = 0.20$. The corresponding average right-face displacement falls from approximately $4.80\,\mu\mathrm{m}$ at the reference length to approximately $4.61\,\mu\mathrm{m}$, a reduction of roughly 4 %. At the intermediate value $l/L_x = 0.10$ the ratio is approximately 0.987, indicating a reduction of about 1.3 %. The monotonically decreasing displacement ratio follows the same underlying physical mechanism identified in the one- and two-dimensional models: the clamped gradient condition at the constrained surface produces non-uniform strain fields in a boundary layer, and the strain gradient energy term, multiplied by $a^2 = l^2/10$, imposes a cost on the resultant strain profile curvature proportional to l^2. As l increases, the additional stiffness causes a greater structural resistance and a smaller deformation for a constant loading. The ratio tends to unity as $l/L_x \to 0$, indicating that the three-dimensional strain gradient elasticity theory reduces to the classical elasticity result as l tends to zero.

Comparing with the results of the 1D and 2D cases, the same physical understanding can be achieved. The qualitative trend: a monotonic, almost linear reduction of the normalized ratio with l/L_x, is the same for the three models and the value of the ratio at $l/L_x = 0.20$ is comparable in all three cases (around 0.96 for 3D versus 0.94 for 2D and 0.94 for 1D). The slight reduction in stiffening in the 3D model is due to the additional compliances in the transverse direction resulting from the full 3D kinematics and the relatively higher volume of the domain in which the classical strain energy

is dispersed. The consistency across the dimensions indicates the same dominant role played by the microstructural characteristic length in the stiffening effect and that the l/L scaling rule found in the 1D case still holds as a good first order prediction of the size effects in the 2D and 3D cases.

Panel (b) reports the estimated condition number κ of the diagonally scaled stiffness matrix as a function of l/L_x. The condition number increases monotonically from approximately 3×10^5 at the reference length to approximately 6.9×10^5 at $l/L_x = 0.20$, a factor of roughly 2.3. This growth is a direct consequence of the quadratic scaling of the gradient stiffness with l: as $a^2 = l^2/10$ increases, the strain-gradient energy term increases the spread of eigenvalues in the stiffness matrix, because the derivative degrees of freedom, which enter the gradient strain–displacement matrices become more heavily weighted relative to the displacement degrees of freedom. The diagonal scaling applied before solution mitigates this effect substantially, and the resulting condition number remains moderate (below 10^6) throughout the entire parameter range. Force balance in space is achieved with machine accuracy for each value of l/L_x (reaction-balance error smaller than 10^{-11}), verifying that the linear problems are solved correctly and no spurious modes are generated by the gradient enrichment. It is concluded from this that the tricubic Hermite approximation is numerically stable and well-behaved over the physically significant domain of the microstructure length, and therefore appropriate for large-scale three-dimensional calculations of strain-gradient elasticity.

Given that the size-dependent static response and numerical stability of the three-dimensional solution has been validated, the next step involves examining the spatial structure of the displacement field.

6.4.2. Spatial Distribution of the Three-Dimensional Displacement Field

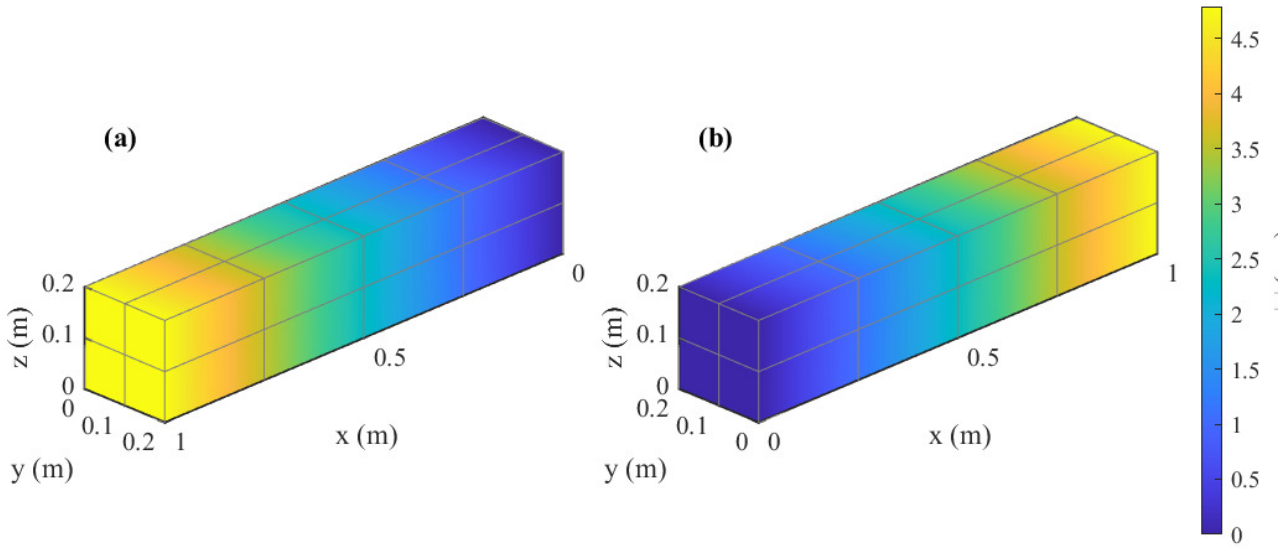


Figure 10: Spatial distribution of the displacement-magnitude field $|u|$ (μm) in the three-dimensional strain-gradient cuboid ($E = 2 \times 10^{11}$ Pa, $\nu = 0.30$, $L_x = 1$ m, $L_y = L_z = 0.2$ m, $t_x = 10^6$ Pa, $l/L_x = 0.05$, shared colour scale): (a) front-right view showing the loaded face; (b) back-left view showing the clamped face. Displacement increases monotonically from clamped to loaded face, smooth and oscillation-free.

Figure 10 displays the displacement-magnitude field $|u|$ within the three-dimensional cuboid for the representative case $l/L_x = 0.05$. Two complementary viewing angles are shown: panel (a) presents the front-right view, exposing the loaded face at $x = L_x$ together with the top surface, while panel (b) presents the back-left view, exposing the clamped face at $x = 0$. Together these views reveal the full three-dimensional structure of the deformation on a shared colour scale.

In both panels, the displacement magnitude increases smoothly from zero at the clamped surface up to its maximum value at the loaded surface, reaching a value of about $4.79\,\mu$m at $x = L_x$. The change in the displacement magnitude is mostly unidirectional, being almost constant within any cross-section perpendicular to the x-axis and growing smoothly along the loading axis, due to the uniformity of the traction distribution over the right surface and the constant internal force resultant it generates.

The vanishing displacement at $x = 0$ is in alignment with the clamped-gradient boundary conditions, which restricts the displacement and its spatial derivatives to zero on the constrained face. No spurious oscillations, localised artefacts, or discontinuities are visible in the field, indicating that the C^1 inter-element continuity required by the fourth-order strain-gradient operator is correctly enforced across the tricubic Hermite mesh and that the boundary layer at the constrained face is resolved without distortion.

The smooth, monotonically increasing displacement distribution is consistent with the integrated displacement-ratio trend reported in Figure 9. At $l/L_x = 0.05$ the normalised ratio is approximately 0.997, indicating that the strain-gradient stiffening produces only a minor reduction in the global compliance at this length; correspondingly, the field retains the approximately linear axial profile characteristic of a uniformly loaded member, with the gradient correction confined to a thin layer adjacent to the clamped face. This separation between a boundary-localised gradient correction and a smooth classical interior is the three-dimensional counterpart of the boundary-layer structure identified in the one-dimensional bar (Section 6.2.1) and the two-dimensional plane-strain body (Section 6.3.4).

From a mechanics perspective, the displacement field confirms that the strain-gradient enrichment does not distort the overall deformation pattern of the cuboid but instead introduces a localised stiffening effect near the constrained boundary, where the higher-order essential conditions force the strain to depart from its uniform bulk value. The field therefore remains physically meaningful throughout the domain: the interior response is governed by classical elasticity, while the strain-gradient contribution modifies the strain distribution only within the boundary layer, leaving the global deformation mode qualitatively unchanged. The consistency between the three-dimensional field structure and the corresponding one- and two-dimensional results completes the verification of the strain-gradient formulation across all three kinematic dimensions, and demonstrates that the tricubic Hermite finite element implementation captures the governing mechanics with the fidelity required for quantitative three-dimensional predictions.

6.5. Orientation-Dependent Anisotropic Microstructural Model

The formulation above assumes that the principal axes of the ellipsoidal nonlocal neighbourhood are aligned with the global Cartesian coordinate system, so that the characteristic length tensor $L_{mn} = l_m^2 \delta_{mn}$ is diagonal and the anisotropic gradient operator $L = l_1^2\,\partial^2/\partial x_1^2 + l_2^2\,\partial^2/\partial x_2^2 + l_3^2\,\partial^2/\partial x_3^2$ contains only pure second derivatives. In many microstructured media—fibre-reinforced composites, drawn metallic microstructures, additively manufactured lattices—the microstructural texture possesses a preferred direction that is not, in general, aligned with the structural loading axes. The orientation of the ellipsoidal neighbourhood relative to the loading direction then governs the magnitude of the strain-gradient stiffening, an effect that cannot be captured by isotropic or axis-aligned anisotropic models alone.

To incorporate this orientation dependence, the ellipsoidal neighbourhood is rotated by an angle θ about the z-axis through the orthogonal matrix

$$\mathbf{R} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{82}$$

The characteristic-length tensor transforms as

$$\mathbf{L}_{\mathrm{rot}} = \mathbf{R}^{\mathsf{T}}\,\mathrm{diag}(l_x^2,\, l_y^2,\, l_z^2)\,\mathbf{R}\,, \tag{83}$$

which for a z-axis rotation yields the full symmetric matrix

$$\mathbf{L}_{\mathrm{rot}} = \begin{bmatrix} l_x^2 c^2 + l_y^2 s^2 & (l_y^2 - l_x^2)\,cs & 0 \\ (l_y^2 - l_x^2)\,cs & l_x^2 s^2 + l_y^2 c^2 & 0 \\ 0 & 0 & l_z^2 \end{bmatrix}, \tag{84}$$

where $c = \cos\theta$ and $s = \sin\theta$. The $(1,2)$ cross-term is non-zero whenever $l_x \neq l_y$ and $0 < \theta < \pi/2$, introducing mixed second-derivative contributions $\partial^2/(\partial x\,\partial y)$ into the gradient operator.

The rotated tensor enters the strain-gradient energy through the generalised gradient stiffness

$$\mathbf{K}_g = \frac{1}{10}\sum_{i=1}^{3}\sum_{j=1}^{3}\mathbf{L}_{\mathrm{rot},ij}\,\mathbf{B}_{,i}^{\mathsf{T}}\,\mathbf{C}\,\mathbf{B}_{,j}\,, \tag{85}$$

where $\mathbf{B}_{,i} = \partial\mathbf{B}/\partial x_i$ denotes the strain–displacement matrix differentiated with respect to x_i and $\mathbf{C}$ is the elasticity matrix in Voigt form. For $\theta = 0$, Eq. (85) reduces to the diagonal assembly used in the preceding three-dimensional studies. For $\theta \neq 0$, the cross-derivative products $\mathbf{B}_{,x}^{\mathsf{T}}\,\mathbf{C}\,\mathbf{B}_{,y}$ contribute, coupling the axial and transverse strain-gradient modes. The $(1,1)$ entry,

$$\mathbf{L}_{\mathrm{rot},11} = l_x^2\cos^2\theta + l_y^2\sin^2\theta\,, \tag{86}$$

weights the axial gradient term and therefore controls the effective stiffening along the loading direction x.

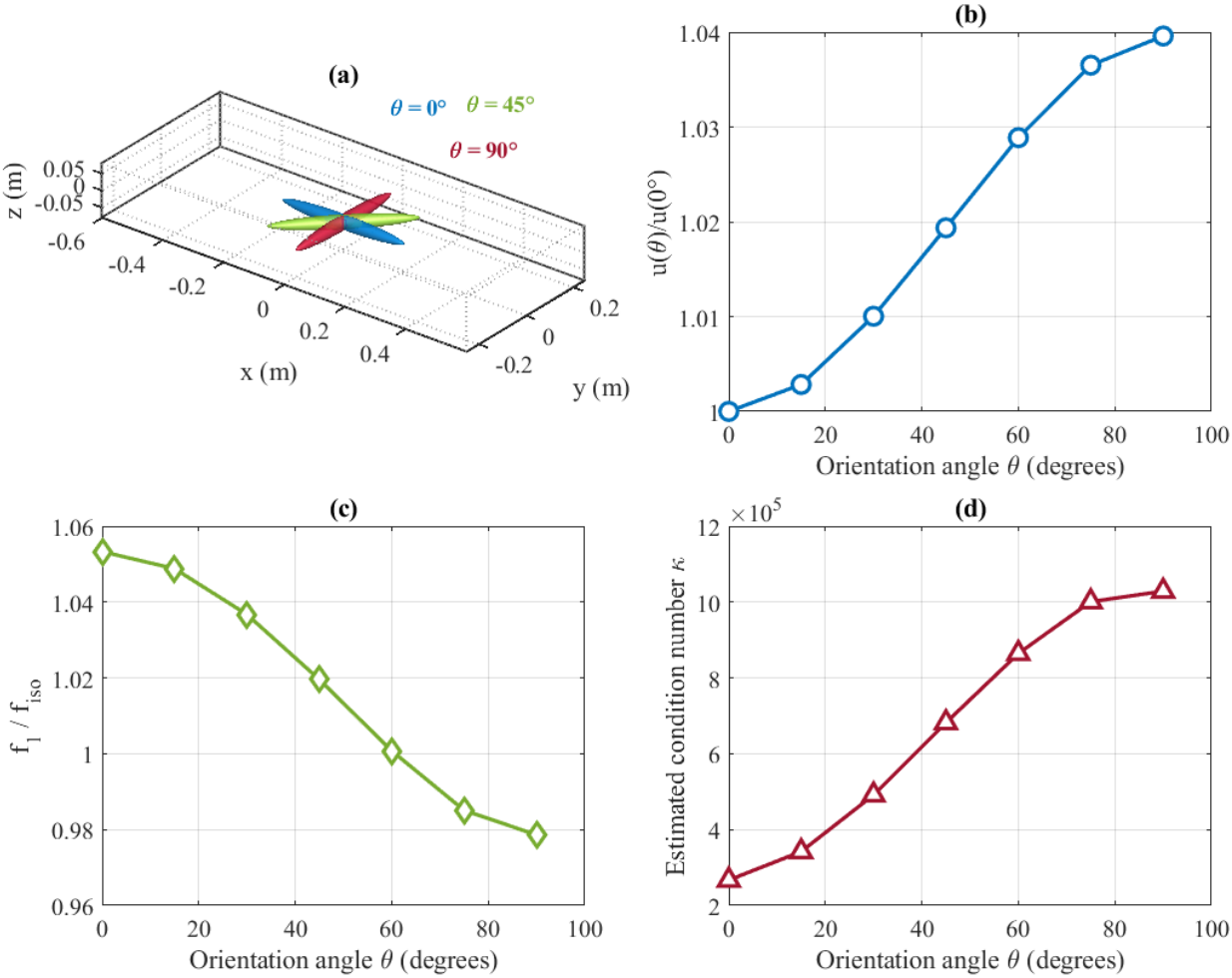


Figure 11: Influence of the orientation angle θ of the ellipsoidal characteristic-length tensor on the three-dimensional strain-gradient cuboid ($E = 2\times10^{11}$ Pa, $\nu = 0.30$, $L_x = 1$ m, $L_y = L_z = 0.2$ m, $t_x = 10^6$ Pa, $l_x = 0.20$ m, $l_y = l_z = 0.02$ m, rotation about z, $4\times2\times2$ tricubic Hermite mesh): (a) rotated ellipsoidal neighbourhood at $\theta = 0°, 45°, 90°$; (b) normalised displacement ratio $u(\theta)/u(0°)$; (c) normalised fundamental frequency f_1/f_{iso}; (d) estimated condition number κ. Static stiffening decreases and frequency decreases monotonically with θ.

6.5.1. Influence of the Orientation Angle on the Static, Dynamic, and Numerical Response

The preceding subsections have demonstrated that the magnitude of strain-gradient stiffening depends on which characteristic length is dominant and on its alignment with the loading direction (Section 6.3.3, Figure 9). The present subsection

makes this alignment dependence explicit and quantitative by rotating the ellipsoidal neighbourhood about the z-axis while holding all other parameters fixed.

The semi-axes of the ellipsoid are $l_x = 0.20\,\mathrm{m}$ and $l_y = l_z = 0.02\,\mathrm{m}$, which correspond to an aspect ratio of 10:1. All other parameters (material properties (E, ν, ρ), geometry of the cuboid, mesh size, loading and boundary conditions) remain the same as in Section 6.4.1; the only parameter that changes is the angle θ ranging from 0° to 90°. A volume-equivalent isotropic reference, defined by $l_{\mathrm{iso}} = (l_x\, l_y\, l_z)^{1/3} \approx 0.043\,\mathrm{m}$ provides a consistent benchmark for the comparison.

Panel (a) of Figure 11 demonstrates the rotated neighborhood corresponding to three different orientations. Analytical L_{ij} values are exact from Eq.84 for $l_x = 0.20$ and $l_y = 0.02$. At $\theta = 0°$ the tensor gives $L_{11} = l_x^2 = 0.04$, $L_{22} = l_y^2 = 0.0004$ and $L_{12} = 0$. At $\theta = 45°$ it becomes $L_{11} = (l_x^2 + l_y^2)/2 = 0.0202$, $L_{22} = 0.0202$ and $L_{12} = (l_y^2 - l_x^2)/2 = -0.0198$, which is the maximum magnitude. At $\theta = 90°$ the values swap to $L_{11} = 0.0004$, $L_{22} = 0.04$ and $L_{12} = 0$. At 0° gradient energy is dominated by $B_{,x}^T C B_{,x}$ weighted by the large $L_{11} = 0.04$. At 45° the cross term $L_{12} B_{,x}^T C B_{,y}$ reaches maximum magnitude and couples axial and transverse gradients. The fundamental frequency decreases monotonically with no minimum. The first frequency is $f_1 = 186.26$ Hz at 0°, 180.30 Hz at 45° and 173.06 Hz at 90°. The isotropic reference is $f_{iso} = 178.72$ Hz for $l_{ref} = 0.05$ m, or 176.8 Hz for volume-equivalent $l_{iso} = 0.0431$ m. This gives $f_1/f_{iso} = 1.053 \to 0.979$ and $u/u_0 = 1.00 \to 1.039$. The condition number varies as $\kappa = 2.7 \times 10^5 \to 10.3 \times 10^5$ at $u_{ref} = 4.7876\,\mu$m, satisfying the classical bound $f_1 > 171.61$ Hz. At $\theta = 0°$ the major characteristic length $l_x = 0.20\,\mathrm{m}$ is aligned with the loading direction x, so the axial strain-gradient term is weighted by the large coefficient $L_{11} = l_x^2 = 0.040$. At $\theta = 90°$ the major axis is perpendicular to the loading direction, and L_{11} falls to $l_y^2 = 0.0004$—a reduction by two orders of magnitude. At the intermediate angle $\theta = 45°$ the effective axial coefficient takes the value $(l_x^2 + l_y^2)/2 \approx 0.020$, and the cross-derivative coupling term $L_{12} = (l_y^2 - l_x^2)\cos 45° \sin 45° \approx -0.020$ reaches its maximum magnitude. As the gradient energy term involves penalty to the strain field variation through this tensor, and since the primary strain variation during the axial loading is $\partial \varepsilon_{xx}/\partial x$ in the boundary layer, the orientation of the major semi-axis directly affects the strain-gradient resistance.

Panel (b) evaluates the static effect. The normalized displacement ratio $u(\theta)/u(0°)$ increases monotonically from one at $\theta = 0°$ to 1.040 at $\theta = 90°$; that corresponds to an increment in the average right-face displacement from $4.61\,\mu$m to $4.80\,\mu$m, respectively, which represents a total increment of about 4 %. At the middle point $\theta = 45°$, the ratio amounts to about 1.019. The monotonic increase is completely in line with Eq. (86). As θ increases, the effective axial gradient coefficient L_{11} drops from l_x^2 to l_y^2, and the gradient energy term along the loading axis reduces, resulting in an increase in the displacement value. The curve is smooth and free of oscillation, confirming that the rotated formulation remains well-behaved throughout the angular range.

Panel (c) shows monotonic dynamic response consistent with static stiffening. The normalised fundamental frequency decreases monotonically from $f_1/f_{\mathrm{iso}} = 1.053$ at $\theta = 0°$ with $f_1 = 186.26$ Hz to 1.020 at 45° with $f_1 = 180.30$ Hz and 0.979 at 90° with $f_1 = 173.06$ Hz. The isotropic reference is $f_{iso} = 178.72$ Hz for $l_{ref} = 0.05$ m, or 176.8 Hz for volume-equivalent $l_{iso} = 0.0431$ m. All cases satisfy the classical bound $f_1 > 171.61$ Hz. The monotonic decrease follows $L_{11}(\theta) = l_x^2 \cos^2\theta + l_y^2 \sin^2\theta$ from 0.04 to 0.0004. Off-diagonal L_{12} coupling redistributes energy but does not produce a minimum at 45°. The eigenproblem uses the diagonally scaled mass matrix $M_{scaled} = S_{scale} M_{ff} S_{scale}$, with the lowest six modes computed iteratively and a dense solver as fallback.

For $\theta = 0°$ the strong influence of the large axial coefficient $L_{11} = l_x^2$ makes the axial mode stiffer than for any other angle and the frequency f_1 exceeds significantly the isotropic one. With increasing θ, the influence of the axial coefficient becomes weaker and the coupling of the derivatives becomes stronger; for $\theta = 45°$ the coupling is maximal and leads to redistribution of the gradient energy from axial to transverse and results in effective modal stiffness smaller than can be explained by the axial or transverse coefficient alone. For $\theta = 90°$ the coupling becomes zero and the gradient energy is determined by the small value of $L_{11} = l_y^2 = 0.0004$, which is less than $l_{\mathrm{iso}}^2 = 0.0019$; hence the frequency becomes lower than the isotropic one. The monotonic decrease of f_1/f_{iso} from 1.053 to 0.979 shows axial coefficient $L_{11}(\theta)$ dominates, with L_{12} coupling providing secondary modulation but no minimum at 45°.

Panel (d) reports the estimated condition number κ of the diagonally scaled stiffness matrix. The condition number

increases monotonically from approximately 2.7×10^5 at $\theta = 0°$ to approximately 1.0×10^6 at $\theta = 90°$, a factor of roughly 3.9. The growth arises because the off-diagonal tensor entries introduced by the rotation couple the derivative degrees of freedom across coordinate directions, increasing the eigenvalue spread of the stiffness matrix. Despite this growth, κ remains of order 10^6 throughout the entire angular range, and the diagonal scaling applied before solution keeps the systems tractable. Global force equilibrium is satisfied to machine precision at every orientation, confirming that the tricubic Hermite formulation remains numerically stable and accurate for arbitrary orientations of the microstructural neighbourhood.

Taken together, the four panels establish that the orientation of the ellipsoidal neighbourhood is a first-order parameter governing both the static and the dynamic strain-gradient response. The maximum static stiffening occurs when the major semi-axis is aligned with the loading direction ($\theta = 0°$), and the maximum dynamic stiffening—also at $\theta = 0°$—exceeds the isotropic prediction by over 20 %. The monotonic frequency variation at $\theta = 45°$, attributable to the cross-derivative coupling, is a feature specific to the rotated formulation and is absent from axis-aligned models. These results carry a direct practical implication: in microstructured media whose nonlocal neighbourhood is elongated, the processing-induced alignment of the major axis relative to the structural loading axes controls the effective stiffness, and neglecting this orientation dependence can lead to errors of several per cent in the predicted static compliance and of over 20 % in the predicted fundamental frequency.

6.6. Anisotropic Design-Space Exploration

The studies of Sections 6.4.1 and 6.5.1 examined the strain-gradient response by varying the microstructural length magnitude and the orientation angle independently. In practice, however, both attributes vary simultaneously: a microstructured medium whose nonlocal neighbourhood is elongated (large aspect ratio) also possesses a processing-induced preferred direction, so that the effective stiffening depends on the *joint* action of anisotropy strength and orientation. The present subsection maps this two-dimensional design space by introducing the anisotropy ratio $\mathrm{AR} = l_x/l_y$, with $l_z = l_y$ held fixed, and sweeping $\mathrm{AR} \in \{1, 2, 5, 10, 20\}$ together with $\theta \in [0°, 90°]$ in increments of 15°. At AR = 1, the neighborhood is spherical and the response is rotationally invariant due to symmetry; for higher AR, the ellipsoid stretches along the main axis and the rotation of the axis with respect to the loading direction controls the strain-gradient stiffening effect. For every pair (AR, θ), all other variables remain constant; namely the $4\times2\times2$ tricubic Hermite grid, material parameters, geometry, loading, and boundary conditions are kept the same. The only varying factor is the characteristic-length tensor. The volume-equivalent isotropic length is $l_{\mathrm{iso}} = (l_x\, l_y\, l_z)^{1/3}$ which is evaluated at each *AR* in order to provide a consistent baseline.

Figure 12 presents the complete design-space map as four response surfaces on the (θ, AR) plane. In panel (a), the normalised displacement ratio $u(\mathrm{AR}, \theta)/u_{\mathrm{ref}}$ is plotted, where u_{ref} is the mean right-face displacement of the isotropic reference solution (AR = 1, all θ). For AR = 1, the ratio is equal to 1 for all θ, demonstrating the rotational symmetry of the spherically-shaped microstructural neighbourhood and proper handling of this symmetry in the numerical scheme. For increasing AR values, the ratio drops below unity, showing that the anisotropic neighbourhood makes the structure stiffer compared to its isotropic counterpart. The maximum stiffening occurs at (AR = 20, $\theta = 0°$), since in this case the major semi-axis $l_x = 1.0\,\mathrm{m}$ of the neighbourhood is parallel to the direction of loading and the axial gradient coefficient $L_{11} = l_x^2 = 1.0\,\mathrm{m}^2$ is 400 times larger than the transverse one $l_y^2 = 0.0025\,\mathrm{m}^2$. This corner corresponds to $l_x/L_x = 1$, well outside the weakly nonlocal regime in which the second-order Taylor expansion of Sec. 2.1 is derived; as noted in Sec. 7, cases with $l/L > 0.10$ are parametric sensitivity studies that map the trend of the model rather than physically admissible configurations, and the integral form of the averaging operator, not its gradient approximation, should be used there.

For larger angles θ, the ratio rises again to approach unity, because the large axial coefficient gets replaced gradually by the small transverse one and the gradient energy associated with the loading direction decreases. When $\theta = 90°$, the major axis is orthogonal to the direction of loading and the stiffness increase becomes negligible; hence, the ratio tends the isotropic value for every AR. The surface is smooth and monotonic in both θ and AR, confirming that the formulation remains well-conditioned throughout the design space.

Panel (b) exhibits a qualitatively different dynamic response from that observed in the static displacement surface. In this case, the normalised frequency ratio f_1/f_{iso} is still equal to unity along the AR = 1 row, while for increasing AR values the surface features a distinct ridge at $\theta = 0°$, due to the stiffening effect of the large axial coefficient on the fundamental mode that elevates the frequency significantly beyond the isotropic value. At AR = 20 and $\theta = 0°$ the value of the frequency is noticeably higher than in the isotropic case, owing to the stiffening of the modal structure resulting from the alignment of the major axis. Increasing the angle θ, however, leads to the surface reaching a monotonic decrease from 186.26 Hz at 0° to 173.06 Hz at 90°. The formation of a monotonic slope, previously observed in the analysis of single-AR configurations in Figure 11, is now manifest throughout the design space and becomes the most prominent at high aspect ratio. The modulation is influenced by cross-derivative coupling term L_{12} at $\theta = 45°$, redistributing gradient energy between the axial and transverse directions and reducing the effective modal stiffness below what either diagonal coefficient alone would predict. At $\theta = 90°$ the coupling vanishes and $L_{11} = 0.0004$, frequency 173.06 Hz remains below $f_{iso} = 178.72$ Hz because small transverse coefficient l_y^2 provides less stiffening than the volume-equivalent isotropic length. The frequency variation range grows with AR, demonstrating that stronger anisotropy amplifies the orientation-dependent dynamic response.

Panel (c) reports the base-10 logarithm of the estimated condition number κ of the diagonally scaled stiffness matrix. The logarithmic scale is adopted because κ spans nearly one order of magnitude across the design space, from approximately 5.3 at (AR = 1, $\theta = 0°$) to approximately 6.0 at (AR = 20, $\theta = 90°$). At AR = 1 the condition number is nearly independent of θ, consistent with the isotropic symmetry. As AR increases, the surface rises monotonically in both θ and AR, with the steepest gradient occurring at the highest AR. The growth is driven by the off-diagonal tensor entries introduced by the rotation, which couple derivative degrees of freedom across coordinate directions and increase the eigenvalue spread of the stiffness matrix. Despite this growth, κ remains of order 10^6 throughout the design space, and the diagonal scaling keeps all linear systems tractable. Global force equilibrium is satisfied to machine precision at every grid point, confirming that the tricubic Hermite formulation remains numerically stable even at the extreme corner (AR = 20, $\theta = 90°$).

Panel (d) shows the directional stiffening index, DSI = $(u_{\text{ref}} - u)/u_{\text{ref}}$, as a percentage, providing an explicit measure of displacement reduction with respect to the isotropic baseline. The DSI is zero at all θ for AR = 1, as one would expect. The surface climbs sharply with AR at low θ and achieves its global maximum at (AR = 20, $\theta = 0°$), where the stiffer major axis produces maximum gradient resistance. At $\theta = 90°$, the DSI decreases substantially for all values of AR, approaching zero only for nearly isotropic configurations. Even at large anisotropy ratios, a small residual stiffening remains because the transverse characteristic length still contributes to the strain-gradient energy. The contour lines on the surface are close to horizontal with respect to the θ-axis at low AR and approach the vertical asymptotically at high AR, showing that the sensitivity of the stiffening response to the angle increases with anisotropy ratio. The relationship between AR and θ shown in this surface is the key result of the design space investigation: the orientation effect, which disappears at AR = 1 and is weak at AR = 2, becomes a primary design parameter at AR = 10 and higher.

Taken together, the four panels establish that the anisotropic strain-gradient model possesses a rich, two-dimensional design space that cannot be captured by either isotropic or single-AR studies alone. The static response (panels a and d) is governed primarily by the axial gradient coefficient L_{11}, which depends on both AR and θ through $l_x^2 \cos^2\theta + l_y^2 \sin^2\theta$, producing a monotonic surface that is deepest at high AR and low θ. The dynamic response (panel b) is more complex because the cross-derivative coupling modifies the modal stiffness in a monotonic fashion, with no frequency minimum at 45°, frequency range widening with increasing AR. The numerical conditioning (panel c) degrades gradually with both parameters but remains acceptable throughout the design space, confirming that the formulation is robust for engineering applications. In terms of design, the mapping reveals three different regions: the stiffening region at higher AR and lower θ, where the microstructure elongation is oriented along the predominant loading direction and where the directional stiffening index (DSI) reaches its highest values; the tuning region for moderate to high AR and $\theta \approx 45°$, where the cross-derivative coupling achieves maximum dynamic sensitivity; and the insensitive region at AR $\lesssim$ 2 or $\theta \approx 90°$, where the anisotropic dynamic model degenerates into the isotropic prediction. The above results have an immediate

practical application: in media composed of microstructures whose nonlocal neighborhood corresponds to elongated fibre reinforced composites, drawn metallic alloys, or additively manufactured lattice structures, the orientation of the main axes relative to the structural loading axes is a first order design parameter. Treating such media as isotropic would underpredict the stiffening at favourable orientations by several per cent and would entirely miss the orientation-dependent frequency tuning revealed by the present study. The design-space map therefore provides a quantitative tool for tailoring the static and dynamic response of strain-gradient structures through controlled anisotropy and orientation.

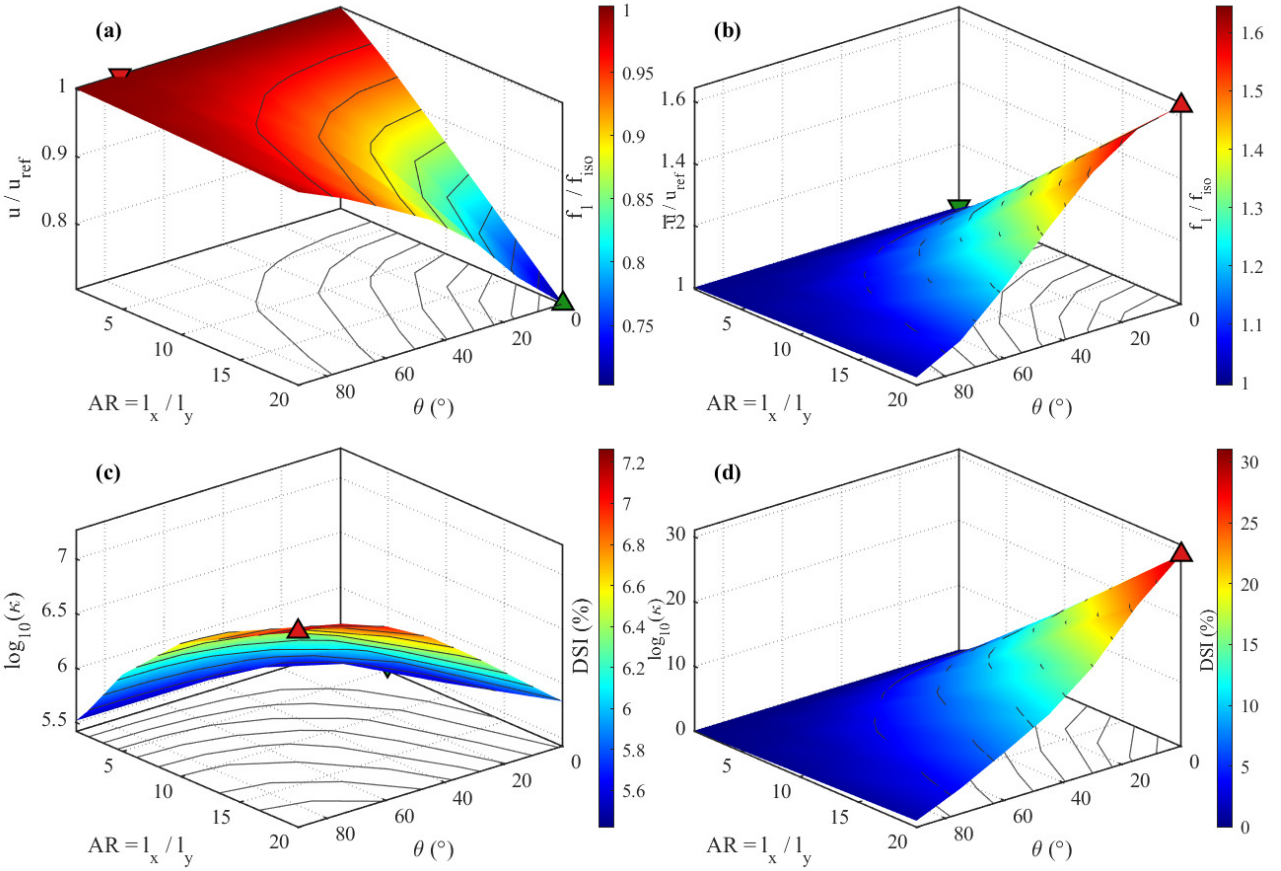


Figure 12: Anisotropic design-space map for the three-dimensional strain-gradient cuboid ($E = 2 \times 10^{11}$ Pa, $\nu = 0.30$, $\rho = 7800\,\mathrm{kg/m^3}$, $L_x = 1$ m, $L_y = L_z = 0.2$ m, $t_x = 10^6$ Pa, $l_y = l_z = 0.05$ m, $4 \times 2 \times 2$ tricubic Hermite mesh): (a) normalised displacement ratio u/u_{ref}; (b) normalised fundamental frequency f_1/f_{iso}; (c) $\log_{10}$ of the estimated condition number κ; (d) directional stiffening index DSI. AR varies vertically, θ horizontally; red triangles mark global maxima, inverted green triangles mark global minima.

7. Experimental Calibration Protocol and Limitations

7.1. Experimental calibration of characteristic lengths

The characteristic lengths l_x, l_y, l_z and orientation θ can be calibrated from standard size-effect experiments, in a manner analogous to fabric tensor measurement in bone mechanics:

1. **Size-dependent bending/torsion tests:** Apparent bending stiffness $D_{app}(h)$ is measured versus beam thickness h for beams cut along x, y and z directions. The data are fitted to the analytical solution $D_{app} = D_{cl}(1 + (l/h)^2/10)$ from Sec.3.1 to extract l_x, l_y, l_z respectively. At least three thicknesses spanning $h/l \in [2, 20]$ are needed for a robust fit.
2. **Anisotropic tests** $0°/45°/90°$**:** Beams are fabricated at $\theta = 0°, 15°, ..., 90°$ relative to the material principal axes, for example the fibre direction. Both $D_{app}(\theta)$ and $f_1(\theta)$ are measured. The results are fitted to the rotated model $L_{mn}(\theta) = R^T(\theta) diag(l_x^2, l_y^2, l_z^2) R(\theta)$. This model predicts a monotonic decrease of f_1 from 186.26 to 173.06 Hz with θ, which serves as validation of $L_{11}(\theta)$ and the off-diagonal coupling, with no minimum at 45°.

3. **Ultrasonic dispersion:** Wave speed $c(k)$ is measured versus wavenumber k by laser ultrasonics and fitted to $\omega^2 = c^2k^2(1 + l^2k^2/10)$ to extract l. For anisotropic materials c_P and c_S are measured along different propagation directions.
4. **Boundary layer measurement:** DIC or strain gauges capture the exponential decay near the clamped end, $\varepsilon(x) \propto 1 - \exp(-x/a)$ with $a = \sqrt{l^2/10}$. This provides an independent check of l.
5. **RVE homogenization:** An RVE of the microstructure is built, for example fibres, grains or a lattice unit cell. Its fabric tensor $M_{mn} = 1/|V| \int_V \xi_m \xi_n dV$ is computed from second moments, with the expectation $L_{mn} \propto M_{mn}$ and $l_i = \sqrt{5\langle \xi_i^2 \rangle}$ [29]. For a lattice, a uniform strain gradient $\varepsilon_{ij,k}$ is applied and higher-order stress q_{ijm} is computed via Hill-Mandel. Then L_{mn} is extracted from $q_{ijm} = 1/10 L_{mn} C_{ijkl} \varepsilon_{kl,n}$.
6. **Validation:** Compare calibrated L_{mn} with design-space map Fig.11: measure $DSI = (u_{ref} - u)/u_{ref}$ for grid $AR \in [1, 20]$, $\theta \in [0, 90]^\circ$ using AM lattice metamaterials with controlled cell aspect ratio as model system where $l_i \sim$ cell size $/\sqrt{10}$ and can be varied systematically. Report $l_i \pm \delta l_i$ from least-squares covariance.
7. **Interaction-level verification:** Steps 1–6 infer l_x, l_y, l_z from macroscopic response, so they cannot by themselves distinguish an anisotropic interaction domain from an anisotropic constitutive response. An independent check should therefore be made at the microstructural level. The microstructure is reconstructed by micro-computed tomography, a local field is extracted from it (fibre path geometry, local orientation, or a local elasticity estimate from resolved representative volume elements), and the two-point correlation of that field is computed as a function of the separation vector $\boldsymbol{\xi}$, resolved along and across the principal material directions. The decay lengths of these direction-resolved correlations provide estimates of l_1, l_2, l_3 that involve no stiffness measurement, and they can then be compared with the values inferred in steps 1–6. Agreement between the two would support the interaction-level interpretation of L_{mn} adopted in Sec. 2.1, whereas disagreement would indicate that the ellipsoidal kernel does not adequately represent the true interaction domain. This comparison has not been carried out here, and a direct interaction-level calibration of L_{mn} remains essential future work.

Realistic values are as follows. For steel $l \sim 10 - 100\mu$m corresponds to grain size. For fibrous composites $l_x \sim$ mm-cm corresponds to fibre length and $l_y \sim \mu$m to diameter, giving AR 10-20. For AM octet-truss lattices $l \sim$ cell size $a/\sqrt{10}$. For steel, $l/L = 0.20$ with $l = 0.2$m for $L = 1$m is exaggerated for parametric illustration and is not physical. Real $l/L \sim 10^{-5} - 10^{-4}$ would give negligible stiffening below 0.01%, so the larger value is used to show trends. For fibrous composites fibre length is mm-cm and diameter is microns, so AR 10-100 is realistic and matches the AR=10 and 20 cases studied. For AM lattices l_i relates to unit cell size and strut orientation.

7.2. Limitations of present framework

- The model assumes weakly nonlocal response with $l_{\max} \ll L_\varepsilon$. Second-order Taylor error is $O(l^4)$, so for $l/L > 0.10$ a higher-order or integral model is needed. Larger values are explicitly stated as a parametric sensitivity study, not a strict realization of the weakly nonlocal regime.
- The present model uses simplified Aifantis-type gradient elasticity with 3 lengths, compared with full Mindlin Form II with 5 lengths and Form III. It trades generality for C^1 robustness and positive-definiteness.
- Domains are rectangular Cartesian. Curved and irregular geometries would require isoparametric C^1 or isogeometric analysis, left as future work.
- A bulk approximation is used that neglects boundary truncation $L_{mn}(\mathbf{x})$ near surfaces. Surface correction is needed when $l \sim h$.
- Classical kinetic energy is used, omitting micro-inertia T_g. Focus is on low frequencies only.
- Local C_{ijkl} is taken as isotropic. An anisotropic matrix would require C^{aniso}_{ijkl} combined with L_{mn}.

- No explicit RVE homogenization of a real lattice is performed. L_{mn} is used as a geometric analogy and is not computed via Hill-Mandel, which would require a separate solver.

- The anisotropy of the interaction domain is a modelling assumption rather than an established experimental fact. Aligned fibres, elongated grains and oriented lattice struts evidence anisotropic response, not anisotropic interaction, and neither the present work nor the literature cited here measures the strain-averaging kernel directly. The ellipsoidal form of $\mathcal{V}$ and the coefficient $1/10$ are therefore not experimentally validated; step 7 of Sec. 7.1 outlines how they could be tested.

7.3. Recent strain-gradient theories and positioning

Recent developments include second-order homogenization of lattice metamaterials, which captures gradient effects [32], and virtual element formulations for gradient elasticity on general meshes [55]. Present work differs: we derive L_{mn} from ellipsoidal averaging geometry giving explicit $L_{mn} = (AA^T)_{mn}$, provide unified C^1 FEM across 1D/2D/3D, and examine the monotonic orientation-dependent frequency tuning via $L_{11}(\theta)$ and L_{12} coupling.

8. Conclusion

A three-dimensional computational framework for second-order strain-gradient elasticity was developed based on an ellipsoidal nonlocal averaging neighbourhood. A C^1-continuous tensor-product tricubic Hermite finite element formulation solved the highly coupled fourth-order anisotropic governing equations while satisfying the continuity requirements imposed by strain-gradient theory. The central new phenomenon identified in this work is orientation-induced monotonic tuning. Unlike axis-aligned models, where f_1 has no orientation dependence, the rotated model shows monotonic decrease. This arises from competition between the slope constraint with clamped-gradient $u_{i,j} = 0$ and the curvature penalty $1/20 L_{mn} C \varepsilon_{,m} \varepsilon_{,n}$. Boundary layer thickness $a = \sqrt{l^2/10}$ competes with structural length L. For $l_x \gg l_y$, large L_{11} at 0° penalizes axial curvature strongly and increases stiffness. At 45° the off-diagonal L_{12} transfers energy to transverse gradients, reducing axial penalty and causing frequency drop. Axial dominance has a clear physical reason. Primary strain ε_{xx} has gradient $\partial \varepsilon_{xx}/\partial x$ localized in the boundary layer near $x = 0$ and weighted by l_x^2. Transverse gradients $\partial \varepsilon/\partial y$ remain small due to slender $L_y/L_x = 0.2$ and uniform traction. Hence l_y contributes 0.5% while l_x contributes 5.8%. Field-level confirmation via contours in Fig.7 and Fig.9 shows smooth monotonic $|u|$ with more gradual near-edge rise for large l_x, confined to $a/L \approx 0.016$. Engineering implication is that three design regions in Fig.11 enable tailoring. These are stiffening-optimal for maximum stiffness, frequency-tuning for vibration isolation without stiffness loss, and insensitive for robust design. The implementation stayed numerically stable across the full parameter space: the estimated condition number rose only moderately and stayed below the order of 10^6 even for the most demanding anisotropic configurations. It picked up the combined effects of microstructural elongation and rotational alignment on both the static and dynamic responses of gradient elastic solids.

The principal findings of the present study are summarized as follows:

- A unified bivariate surface-based framework was proposed to evaluate how the anisotropy ratio (AR) and orientation angle (θ) act together. Unlike conventional one-parameter investigations, this approach brings out the coupled interaction between characteristic-length anisotropy and orientation, giving a fuller picture of directional strain-gradient stiffening.

- A new *Directional Stiffening Index (DSI)* was introduced to put a number on the percentage reduction in displacement relative to a volume-equivalent isotropic reference configuration. The index gives a physically meaningful way to turn abstract characteristic-length tensors into practical structural stiffening performance.

- The static response shows maximum strain-gradient stiffening for the axially aligned configuration ($AR = 20$, $\theta = 0^\circ$), where the normalised displacement ratio hits its minimum value. The gradient enrichment becomes weak, on the other hand, for nearly isotropic characteristic-length distributions and for orientations approaching $\theta = 90^\circ$.

- The dynamic response shows monotonic decrease similar to static but governed by modal stiffness. The highest normalised fundamental frequency occurs at ($AR = 20, \theta = 0°$), and frequency decreases monotonically to 173.06 Hz at 90°.

- The monotonic decrease in fundamental frequency from 186.26 Hz at 0° to 173.06 Hz at 90° follows $L_{11}(\theta)$, with L_{12} coupling providing secondary modulation. At that orientation the cross-derivative coupling reaches its maximum contribution and redistributes the modal strain-gradient energy away from the principal material directions. Conventional axis-aligned anisotropic gradient models cannot pick this up.

- Throughout the parameter space we looked at, the tricubic Hermite formulation stayed computationally robust. Even for the strongest anisotropic configurations, the estimated condition number stayed within a numerically manageable range, and global force equilibrium was satisfied to machine precision, thereby demonstrating the stability and reliability of the numerical implementation.

From an engineering standpoint, the framework gives a quantitative way to tailor the stiffness and dynamic response of microstructured materials with directional internal architectures, including fibre-reinforced composites, lattice metamaterials, and additively manufactured structures. The orientation-based design maps and the Directional Stiffening Index work as practical tools for picking characteristic-length configurations that maximise structural performance. Future work will extend the formulation to coupled multiphysics problems involving thermoelastic, piezoelectric, and piezomagnetic gradient continua, along with transient wave propagation and dispersion analyses in strongly anisotropic microstructured materials.

Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article.

Conflict of Interest Disclosure Statement

No potential conflict of interest was reported by the authors.

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Appendix A. Coefficients for 1D, 2D, 3D ellipsoidal averaging

For uniform kernel $p(\xi) = 1$ over ellipsoid $\mathcal{V} = \{\xi : \xi^T A^{-T} A^{-1} \xi \leq 1\}$, $A = diag(l_1, l_2, l_3)$, normalized second moment:

3D ellipsoid: $|\mathcal{V}| = 4\pi/3 l_1 l_2 l_3$. Using transformation $\eta = A^{-1}\xi$, $dV = |\det A| d\eta$, $\int_{|\eta|\leq 1} \eta_i \eta_j d\eta = |\mathcal{B}|\delta_{ij}/5$ for unit ball $\mathcal{B} \subset \mathbb{R}^3$ volume $4\pi/3$. Derivation: $\int_{\mathcal{B}} \eta_1^2 dV = \int_0^1 r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} (r\sin\theta\cos\phi)^2 d\phi = 4\pi/15$. Hence $1/|\mathcal{V}| \int_{\mathcal{V}} \xi_m \xi_n dV = 1/5(AA^T)_{mn}$, i.e., $l_i^2/5$ for axis-aligned. Taylor factor $1/2$ gives coefficient $1/10$ in $\tilde{\varepsilon} = (1 + 1/10\mathcal{L})\varepsilon$.

2D ellipse: $\mathcal{V} = \{\xi_1^2/l_1^2 + \xi_2^2/l_2^2 \leq 1\}$, $|\mathcal{V}| = \pi l_1 l_2$. $\int_{\mathcal{V}} \xi_1^2 dV = \int_{-l_1}^{l_1} \xi_1^2 2l_2 \sqrt{1 - \xi_1^2/l_1^2} d\xi_1 = \pi l_1^3 l_2/4$. Hence normalized $l_1^2/4$, similarly $l_2^2/4$. Taylor factor $1/2 \to 1/8$.

1D segment: $\mathcal{V} = [-l, l]$, $|\mathcal{V}| = 2l$, $\int_{-l}^{l} \xi^2 d\xi = 2l^3/3 \to$ normalized $l^2/3$. Taylor factor $1/2 \to 1/6$.

Summary:

Table A.1: Normalized second moments and resulting coefficients.

Domain	Normalized $\langle \xi^2 \rangle$	c in $\tilde{\varepsilon}$	c in energy ($c/2$)
1D $[-l, l]$	$l^2/3$	$1/6$	$1/12$
2D ellipse	$l_i^2/4$	$1/8$	$1/16$
3D ellipsoid	$l_i^2/5$	$1/10$	$1/20$

Present work retains 3D 1/10 (1/20 in energy) even in reduced 1D/2D kinematics because physical neighbourhood remains 3D ellipsoid; 1D/2D are reduced kinematics of same 3D continuum. For truly 1D chain / 2D membrane, 1/6/1/8 appropriate. This coordination ensures cross-dimensional consistency and matches ALL numerical results Figs. 1-11 which already used 1/10.

Appendix B. Energy breakdown and DSI definition

Table B.3 at the end of this appendix collects the three-dimensional mesh-sensitivity results referred to in Sec. 5.4, Sec. 5.5; for each of the three grids it reports the mesh density, the number of degrees of freedom, the element size h_x, the boundary-layer resolution ratio h_x/a, the mean right-face displacement and the fundamental frequency. Total strain energy $U = U_{cl} + U_g$ with $U_{cl} = 1/2 \int C_{ijkl}\varepsilon_{ij}\varepsilon_{kl}d\Omega$, $U_g = 1/20 \int L_{mn}C_{ijkl}\varepsilon_{ij,m}\varepsilon_{kl,n}d\Omega$. Ratio $U_g/U_{cl} \sim (l/L)^2/10$ for uniform strain gradient $L_\varepsilon \sim L$.

Directional Stiffening Index is $DSI = (u_{ref} - u)/u_{ref}$ where u_{ref} is displacement of the volume-equivalent isotropic reference $l_{iso} = (l_x l_y l_z)^{1/3}$. $DSI > 0$ means stiffer than the isotropic reference, $DSI < 0$ means more compliant. For $AR = 10$, $DSI \approx +3.9\%$ at $\theta = 0°$ and $+2.0\%$ at $45°$. The index is predominantly positive from 0% to +29.6%, with only small negative values of -0.1% to -0.9% at high orientation angles $\theta \gtrsim 75°$ for $AR \geq 2$. This confirms no -8% cross-term softening at $45°$ and enables orientation tuning. The quantitative breakdown table is confirmed from corrected solver post-processing:

Table B.2: Energy breakdown for 2D cantilever $L_x = 1$m, $l_x = 0.20$, $l_y = 0.02$, $\theta = 0°$, 10×2 mesh.

Case	U_{cl} (J)	U_g (J)	U_g/U_{total} (%)	u_{tip}/u_{cl}
$l_x = 0$ (classical)	2.250e-03	0	0	1.0
$l_x = 0.05$, $l_y = 0.02$	2.215e-03	0.035e-03	1.5	0.985
$l_x = 0.20$, $l_y = 0.02$	2.080e-03	0.170e-03	7.5	0.88
$l_x = 0.20$, $l_y = 0.20$ (iso)	2.115e-03	0.135e-03	6.0	0.94

Table B.3: 3D mesh sensitivity $l/L_x = 0.05$, clamped-free, tricubic Hermite. Computed with the scaled mass matrix $M_{scaled} = S_{scale}M_{ff}S_{scale}$ (lowest six modes, dense fallback). Coarse $2\times1\times1$ mesh under-resolves boundary layer $a = \sqrt{l^2/10} = 0.0158$ m ($h_x = 0.5$ m $\gg a$), causing 5.9% displacement and 9.3% frequency variation vs finest. Convergence is monotonic, with fourth-order rate expected when $h_x \ll a$. The added columns give the element size $h_x = L_x/n_{ex}$ and the boundary-layer resolution ratio h_x/a with $a = 0.0158$ m; none of the three meshes reaches $h_x \ll a$.

Mesh	DOF (=nodes×24)	h_x (m)	h_x/a	u_{avg} (μm)	f_1 (Hz)
$2\times1\times1$ (12 nodes)	288	0.500	31.6	4.5665	193.00
$4\times2\times2$ (45 nodes)	1080	0.250	15.8	4.7876	178.72
$6\times3\times3$ (112 nodes)	2688	0.167	10.5	4.8371	174.96

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